General designs and an irrationality exponent for Catalan's constant

Marcos Costa Santos Carreira

9 October 2026

Abstract

The preprint "Catalan's constant is irrational" (OpenAI, 24 September 2026) proves the irrationality of Catalan's constant by comparing two bounds for a sequence of determinants built from five integer parameters; the certified gap between the two bounds is about 0.00013. We state four conditions on the parameters under which the argument of the preprint goes through unchanged, verify each step of the preprint for general parameters, and exhibit a second choice of parameters that satisfies the four conditions with a certified gap of 0.0026, by exact computations of the same kind as in the preprint. For this choice we bound all minors of the determinants, by an energy estimate with fewer nodes and by bounds for Schur polynomials, and obtain an upper bound below two thousand seven hundred and fifteen for the irrationality exponent of Catalan's constant; measurements of the determinants near the constant indicate that the method gives about two thousand one hundred, and show which bounds account for the difference. We also show that the best gap the real-place estimate of the preprint can certify for given parameters is the value of an equilibrium problem for a logarithmic energy of two measures, and compute it for both choices. Attempts to extend the method to the Dirichlet beta value at four are reported in a companion note.

1 Introduction

Let $G = \sum_{k\geq 0}(-1)^k(2k+1)^{-2}$ be Catalan's constant. The preprint [1] constructs, for each positive integer N, a determinant Δ_N of size $48N$ whose entries are rational linear combinations of 1 and G. If G were rational, two estimates would hold along the primes $N = p$: a lower bound from the denominators of Δ_p, which are nonzero by an arithmetic certificate,

$$\liminf_{p\to\infty}\Big(\frac{\log|\Delta_p|}{(48p)^2} - \frac{1}{2}\log 2\Big) \geq -2.2908096,$$

and an upper bound from an integral representation, valid without any hypothesis,

$$\limsup_{N\to\infty}\Big(\frac{\log|\Delta_N|}{(48N)^2} - \frac{1}{2}\log 2\Big) \leq -2.2909399.$$

The two are incompatible, so G is irrational. The difference between the constants is 1.3×10^{-4}. With bounds for all complementary minors of Δ_N, Jha [9] turns the same comparison into a finite irrationality exponent, $\mu(G) \leq 10^8$; in that argument the exponent decreases as the gap grows. Litsey [10] obtains $\mu(G) \leq 8\,127$ by a different perturbation estimate and a new certificate that raises the gap of the preprint's design to 1.0×10^{-3}; both bounds are formalized in Lean. A different proof of the irrationality of G, by weighted tails of the defining series, is proposed in [6]; Wachs [7] shows that its final estimate does not follow from its derivation, and Appendix D confirms this by an independent exact computation.

The construction depends on five positive integers per unit of N, which the preprint fixes as $(48, 7, 4, 4, 2)$. A line-by-line check of [1] for this choice, with both finite certificates reproduced, is in [3]. The argument of [1] is also formalised in Lean 4 for this choice [2]; the formalisation fixes the parameters of that design. Here we treat the five integers as free parameters. The contributions are the following.

1. Theorem 6 gives four conditions on a design $(n_0, b_0, q_0, g_0, h_0)$: two inequalities, the invertibility of three fixed rational matrices, and one numerical inequality between the two bounds. Under these conditions the argument of [1] proves that G is irrational. The proof goes through the preprint section by section and records every place where the parameters enter (Section 3).

2. For the design $(96, 15, 6, 8, 5)$ we certify all four conditions (Section 4). The lower constant is -2.2755480, the upper constant is at most -2.2781336, and the certified gap is 2.59×10^{-3}, about twenty times that of the preprint. The certificate for the upper bound follows the method of [1, Sec. 7]: exact derivative numerators, exhaustive isolation of all stationary points, and interval arithmetic for every value.

3. Section 5 makes the comparison quantitative. We check the ingredients of [9] for the design $(96, 15, 6, 8, 5)$ and bound the minors of size $m < n$ by an energy estimate for m nodes that uses the certificate of (C4), in place of padding to size n, and with sharper bounds for the row and column determinants, obtained from Schur polynomials and from a singular-value estimate. With the degree bound of [10], the gap of that design then gives $\mu(G) < 2\,715$ (Theorem 35); for the design of the preprint, [9] obtains 10^8 and [10] obtains $8\,127$. Measurements of the determinants near G indicate that the method gives about 2.1×10^3 (Observation 37); the remaining difference lies in the treatment of the row and column determinants outside the energy and in the bound for the entries of $\Phi^{(1)}$ (Section 5.7).

4. The upper constant of [1, Prop. 6.1] depends on two freely chosen trial sequences. Proposition 45 shows that its infimum over all trial sequences equals the maximum of a logarithmic energy of two probability measures, and that the trial sequences of a good certificate lie close to the moments of the maximising measures (Section 6). Numerically, the best gap the method can certify is between 0.00111 and 0.00138 for the design of the preprint and between 0.00259 and 0.00269 for $(96, 15, 6, 8, 5)$; the tail bases of our certificate are placed at the edges of the supports of the maximising measures.

5. Attempts to carry the method to $\beta(4) = \sum_{k\geq 0}(-1)^k(2k+1)^{-4}$ are reported in a companion note [12]; within the families tested there, the constructions with small denominators tie $\beta(4)$ to a polylogarithmic constant, and the constructions that separate it have much larger denominators.

The design $(96, 15, 6, 8, 5)$ was found by a local search over designs scored by the numerical value of the gap; that search is not part of any proof and is described only briefly in Section 4. All statements labelled Observation are numerical and are not used in the proofs of Theorems 6 and 17. This note has not been refereed.

We follow the notation of [1] and refer to its numbered equations and statements as, for example, [1, Eq. (78)] and [1, Prop. 6.1].

2 Designs and conditions

Definition 1 (Design)**.** A design is a list $(n_0, b_0, q_0, g_0, h_0)$ of positive integers. Its derived integers are

$$a_0 = b_0 + q_0, \quad L_0 = n_0 + a_0, \quad C_0 = L_0 + g_0, \quad H_0 = C_0 + h_0, \quad A_0 = a_0 + 2g_0,$$

and its ratios are

$$\alpha = \frac{a_0}{n_0}, \quad \beta = \frac{b_0}{n_0}, \quad \gamma_q = \frac{q_0}{n_0}, \quad \gamma_g = \frac{g_0}{n_0}, \quad \eta = \frac{h_0}{n_0}, \quad \delta = \frac{q_0 + g_0 + h_0}{n_0}.$$

For a positive integer N the scaled parameters are $n = n_0N$, $b = b_0N$, $q = q_0N$, $g = g_0N$, $h = h_0N$, and a, L, C, H, A are defined from them by the same formulas. The determinant Δ_N of the design is defined in Definition 2 below; for the design of [1], these are the parameters of [1, Eq. (2)].

The preprint uses the design $(48, 7, 4, 4, 2)$, for which $q_0 = g_0$; it writes γ for both γ_q and γ_g.

Definition 2 (Rows, moments and the determinant)**.** Let $f(t) = \sqrt{1 - t^2}$ on $(-1, 1)$, with the positive branch, and $w = t/(1 + f)$; the involution $w \mapsto w^{-1}$, denoted by a star, fixes t and sends f to $-f$. For a design, a positive integer N and $r \geq 0$ put

$$R_r = (1 - t)^h \, t^{C-1} w^{r-g}, \qquad P_r = \frac{R_r + R_r^*}{2}, \qquad D_r = \frac{t\,(R_r^* - R_r)}{2f}$$

[1, Eq. (3)]. With $d = |r - g|$ and the Chebyshev polynomials T_d, U_d, one has $P_r = (1-t)^h t^{C-1} T_d(1/t)$ and $D_r = \operatorname{sgn}(r - g)(1 - t)^h t^{C-1} U_{d-1}(1/t)$, so both are polynomials with integer coefficients. For nonnegative integers i, j let

$$M(i, j) = \int_{-1}^{1}\int_{0}^{1} \frac{|t|\, t^i s^j}{f(t)\,(1 - ts)}\, ds\, dt, \qquad Z(i, j) = \int_{0}^{1}\int_{0}^{1} \frac{t^i s^j}{1 - ts}\, ds\, dt$$

[1, Eq. (6)], extended bilinearly to polynomials in t and in s. Put $\psi_k(s) = s^{b+k}(1 - s)^q$. The determinant of the design is

$$\Delta_N = \det_{0 \leq r,k < n} \Big(M(P_r, \psi_k) - \tfrac{3}{2}\, Z(D_r, \psi_k) \Big)$$

[1, Eq. (7)].

The moments are $M(i, j) = M^0(i, j) + 4G\, c_{(i-j)/2} + \frac{3}{2}\zeta(2)\, c_{(j-i-1)/2}$ and $Z(i, j) = Z^0(i, j) + [i = j]\,\zeta(2)$, where $c_l = 4^{-l}\binom{2l}{l}$ for integers $l \geq 0$ and $c_z = 0$ otherwise, and M^0, Z^0 are explicit rational arrays [1, Eqs. (14), (16), (17)]. The rows are chosen so that $\zeta(2)$ cancels from every entry (Proposition 8); the entries of Δ_N therefore lie in $\mathbb{Q} + \mathbb{Q}G$. With $d\mu(t) = |t|\, dt/f(t)$ on $(-1, 1)$ and $A_r(t) = P_r(t) - \mathbf{1}_{t>0}\, \frac{3}{2}\frac{f(t)}{t}\, D_r(t)$, the entries are also

$$\int_{-1}^{1}\int_{0}^{1} A_r(t)\, \frac{\psi_k(s)}{1 - ts}\, ds\, d\mu(t)$$

[1, Sec. 5.1]. The proof of irrationality compares a lower bound for $|\Delta_p|$ along primes p, which holds if G is rational and comes from the denominators of the entries (Sections 3.2–3.4), with an upper bound from this integral representation (Section 3.5).

Definition 3 (Loss function)**.** For real y put $(y)_+ = \max(y, 0)$. For $x \ge 0$ let

$$d_0(x) = \left(\min(x,\, L_0 - x,\, A_0) - \max(b_0,\, H_0 - 2x)\right)_+ ,$$

and, with $K = L_0 - x$ and $J = H_0 - x$,

$$\begin{aligned} \mathcal{R}(x) &= (K)_+ + (K - A_0)_+ + (J - \max(b_0, K))_+ , \\ \mathcal{S}(x) &= (\min(A_0, K) - b_0)_+ + (\min(b_0, J) - \max(0, K))_+ . \end{aligned}$$

The loss function of the design is

$$\mathcal{D}(x) = \begin{cases} 2n_0 - (d_0(x) - q_0)_+ , & 0 \le x \le H_0/2, \\ \min\left(2n_0,\, n_0 + \mathcal{R}(x),\, 2\mathcal{R}(x) + \mathcal{S}(x)\right), & H_0/2 < x \le H_0, \\ 0, & x > H_0, \end{cases}$$

and its finite-place constant is

$$F_{\mathrm{low}} = -\frac{1}{n_0^2}\int_0^{H_0} \mathcal{D}(x)\,dx - \Big(\frac{1}{2} + \frac{\delta}{2} + \frac{\delta^2}{8}\Big)\log 2.$$

The function $\mathcal{D}$ is piecewise linear with finitely many pieces, so F_{low} is an explicit number of the form $r_1 + r_2 \log 2$ with r_1, r_2 rational. For the design of the preprint, $\int_0^{65} \mathcal{D} = 8609/2$ and $\mathcal{D}$ is the function d of [1, Table (43)].

Definition 4 (Fixed matrices)**.** For $0 \le r \le n_0$ let P_r and D_r be the row polynomials of Definition 2 with $N = 1$, that is with parameters $(n_0, b_0, q_0, g_0, h_0)$; the index $r = n_0$ is the auxiliary row. Let M^0 and Z^0 be the rational arrays above. The fixed matrix B of the design is the rational $(n_0 + 1) \times n_0$ matrix

$$B(r,k) = \sum_{j=0}^{q_0} (-1)^j \binom{q_0}{j} \Big(M^0(P_r,\, b_0 + k + j) - \frac{3}{2} Z^0(D_r,\, b_0 + k + j) \Big), \qquad 0 \le r \le n_0,\ 0 \le k < n_0,$$

and B_0, B_1 are its $n_0 \times n_0$ submatrices formed by the rows $0, \dots, n_0 - 1$ and $1, \dots, n_0$.

For the preprint's design this is [1, Eq. (47)].

Definition 5 (Real-place constant)**.** For $\kappa \in \{1, 2\}$, admissible sequences p, v in the sense of [1, Sec. 6] (that is, $|u_k| \le c\, r^k$ for some $c < \infty$ and $0 < r < 1$) and $\lambda \in \Lambda_\kappa$, where $\Lambda_2 = \{0\}$ and $\Lambda_1 = [0, \infty)$, let

$$U_\kappa(p, v, \lambda) = (-1 + \alpha + \gamma_g)\log 2 + \kappa \|p\|_*^2 + \tfrac{1}{2}\|v\|_*^2 + \sup_{\substack{-1 \le x < 1 \\ x \ne 0}} X_{p,v,\lambda}(x) + \sup_{0 < s < 1} Y_v(s),$$

where, as in [1, Eq. (77)],

$$\|u\|_*^2 = \sum_{k \ge 1} \frac{u_k^2}{k}, \qquad T(u, x) = \sum_{k \ge 1} \frac{u_k T_k(x)}{k}, \qquad S(u, x) = \sum_{k \ge 1} \frac{u_k x^k}{k},$$

and

$$\begin{aligned} X_{p,v,\lambda}(x) &= (\alpha + 2\gamma_g)\log|x| + 2\eta \log(1 - x) - \Big(\frac{\kappa}{2} + \alpha + \eta + \gamma_g\Big)\log(1 + x^2) \\ &\quad + \lambda\Big(2\gamma_g - \frac{2x^2}{1 + x^2}\Big) - 2\kappa T(p, x) - S(v, x), \\ Y_v(s) &= \beta \log s + \gamma_q \log(1 - s) + 2T(v, s). \end{aligned}$$

For the preprint's design this is the right side of [1, Eq. (78)].

Theorem 6. *Let $(n_0, b_0, q_0, g_0, h_0)$ be a design satisfying*

(C1) $2g_0 < n_0 - 1$*;*

(C2) $2b_0 < H_0$*;*

(C3) $\det B_0 \neq 0$*,* $\det(B_0 + B_1) \neq 0$ *and* $\det(B_0 - B_1) \neq 0$*;*

(C4) there are admissible sequences $p^{(\kappa)}, v^{(\kappa)}$ and $\lambda^{(\kappa)} \in \Lambda_\kappa$, for $\kappa = 1, 2$, with

$$\max_{\kappa \in \{1,2\}} U_\kappa(p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}) < F_{\mathrm{low}}.$$

Then Catalan's constant is irrational.

Condition (C2) holds whenever $b_0 \leq n_0$. For the preprint's design (C1) reads $8 < 47$, (C2) reads $14 < 65$, (C3) is [1, Eq. (48)], and (C4) is [1, Props. 3.4 and 7.1].

3 Proof of Theorem 6

Throughout this section $(n_0, b_0, q_0, g_0, h_0)$ is a fixed design, N is a positive integer, and n, b, q, g, h, a, L, C, H, A are the scaled parameters of Definition 1. The proof follows [1] section by section. For each statement we list the steps of the preprint whose validity depends on the design and show that they hold under the stated conditions; every other step is a formal identity or an estimate that does not involve the five parameters, and is used unchanged.

3.1 Rows, moments and rationality

Lemma 7 (Supports)**.** *Assume (C1). For $0 \leq r < n$, the polynomials P_r and D_r of Definition 2 satisfy* $\operatorname{supp} P_r \subseteq \{A, \dots, H-1\}$ *and* $\operatorname{supp} D_r \subseteq \{A+1, \dots, H-1\}$*. For every $r \geq 0$, $tP_r/f - D_r = O(t^{C+r-g})$.*

Proof. With $d = |r - g|$, $P_r = (1-t)^h t^{C-1} T_d(1/t)$ and $D_r = \pm(1-t)^h t^{C-1} U_{d-1}(1/t)$ [1, Sec. 2]. Their lowest powers of t are $C - 1 - d$ and $C - d$, and their highest are at most $C - 1 + h = H - 1$. Since $C - 1 - A = n - 1 - g$, the lower bounds require $d \leq n - 1 - g$ for $0 \leq r < n$, that is $\max(g, n-1-g) \leq n-1-g$, which is $2g \leq n-1$. By (C1), $2g_0 \leq n_0 - 2$, so $2g \leq (n_0 - 2)N \leq n - 2$. The last assertion is [1, Eq. (5)], whose proof uses only the definitions. □

Proposition 8 (Rationality; [1, Prop. 2.1])**.** *For every design and every N, each raw entry $F_j(r)$ with $0 \leq r < n$, $b \leq j < L$ lies in $\mathbb{Q} + \mathbb{Q}G$. If $G \in \mathbb{Q}$, then $\Delta_N \in \mathbb{Q}$.*

Proof. The moment evaluations [1, Eqs. (9)–(19)] do not involve the design. In the proof of [1, Prop. 2.1] the coefficient of $\zeta(2)$ in $F_j(r)$ is $\frac{3}{2}[t^j](tP_r/f - D_r)$, which vanishes for $j < L$ by Lemma 7 and $C + r - g \geq C - g = L$. The filter $s^{b+k}(1-s)^q$, $0 \leq k < n$, uses the raw columns $b \leq j \leq b + n - 1 + q = L - 1$, because $a = b + q$. □

3.2 The prime two

Proposition 9 ([1, Prop. 2.2])**.** *Assume (C1) and $G \in \mathbb{Q}$. Then $v_2(\Delta_N) \geq -(\frac{\delta}{2} + \frac{\delta^2}{8})n^2 - O_G(n \log(n+2))$ for every N, and the same bound holds for every full minor of the raw column matrix.*

Proof. The proof of [1, Prop. 2.2] uses the design only through the following facts: all moment indices lie in $[0, H-1]$ (Lemma 7); $C \geq n$, which holds since $C = n + a + g$; and $H - b = n(1+\delta)$, which is the definition of δ. With these, [1, Eq. (30)] holds verbatim. In its minimisation the expression decreases in l because $l - C \leq n - C \leq 0$, so the minimum over real $m, l \geq 0$ with $m + l \leq n$ is at least the unconstrained minimum $-(\delta/2 + \delta^2/8)n^2$ of $\frac{n^2}{2} - (2+\delta)nm + 2m^2$. □

3.3 Odd primes and the finite-place bound

Proposition 10 ([1, Prop. 3.4])**.** *Assume (C1), (C2) and $G \in \mathbb{Q}$. Along every sequence $N \to \infty$ with $\Delta_N \neq 0$,*

$$\liminf_{N\to\infty} \Big(\frac{\log|\Delta_N|}{n^2} - \frac{1}{2}\log 2 \Big) \geq F_{\mathrm{low}}.$$

Proof. We go through [1, Sec. 3].

Digit reductions. [1, Lemma 3.1] concerns $M^0(i,j)$ and $Z^0(i,j)$ for $0 \leq i, j < H$ and primes $2\sqrt{H} < p \leq H$; it involves the design only through H, and its proof uses only $H < p^2/4$.

Leading layer. In [1, Eqs. (36)–(37)] the degree bounds $\deg(tP\,E) \leq H + p - 1$ and $\deg D < H$ hold by Lemma 7, and $U_\ell = 0$ for $\ell \leq A$ because tP_r and D_r have no terms below degree $A+1$ (Lemma 7). For $2\sqrt{H} < p \leq H/2$ the pairs $(\ell, p+\ell)$ with $\max(b, H-2p) \leq \ell < \min(p, L-p, A)$ consist of raw columns ($b \leq \ell$ and $p + \ell < L$), so [1, Eq. (38)] holds: $v_p(\Delta_N) \geq -2n + (d_0 - q)_+$ with $d_0 = (\min(p, L-p, A) - \max(b, H-2p))_+$.

Central layer and elimination. For $H/2 < p \leq H$, [1, Lemma 3.2] uses only $H - 1 - J = p - 1$ and $j < p$. In [1, Lemma 3.3]: $p > b$ holds because $p > H/2 > b$ by (C2); $J < p$ because $p > H/2$; $K < J$ because $L < H$; the unpaired high columns have leading residue $-V_\ell$ because $\ell < b < A$. The identity $R_0 = R$ of [1, Eq. (41)] is checked in the four cases $K \leq 0$, $0 < K \leq b$, $b < K \leq A$, $K > A$ using only $K < J$; the values are $(J-b)_+$, $K + (J-b)_+$, J and $K + J - A$ in both expressions. The rank argument and the Cauchy–Binet transfer do not involve the design. Hence [1, Eq. (40)] holds.

Loss function. Write $x = p/N$. All quantities in [1, Eqs. (38), (40)] are linear in p and N with integer coefficients from the design, so for odd primes $2\sqrt{H} < p \leq H$ not dividing the denominator of G we obtain $-v_p(\Delta_N) \leq N\,\mathcal{D}(p/N)$ exactly, with $\mathcal{D}$ of Definition 3. Primes $p > H$ give $v_p(\Delta_N) \geq 0$ for large N, and primes $p \leq 2\sqrt{H}$ contribute $o(n^2)$, by the arguments of [1, Sec. 3.4], which do not involve the design.

Prime number theorem. [1, Eq. (45)] states $N^{-1} \sum_{p \leq H_0 N} d(p/N) \log p \to \int_0^{H_0} d$ for its continuous d. The same argument applies to every bounded function with finitely many discontinuities, in particular to $\mathcal{D}$: on a fixed partition the prime number theorem gives $(\theta(Nx_j) - \theta(Nx_{j-1}))/N \to x_j - x_{j-1}$, the weighted sum lies between the lower and upper Darboux sums of the partition up to $o(1)$, and these sums converge to $\int \mathcal{D}$ as the mesh tends to zero because $\mathcal{D}$ is Riemann integrable. The bound $\mathcal{D} \leq 2n_0$ shows that the prime 2 and the primes below $2\sqrt{H}$ change the sum by $o(1)$.

Conclusion. Adding Proposition 9 and using $\log|\Delta_N| = \sum_p v_p(\Delta_N) \log p$, $n^2 = n_0^2 N^2$,

$$\frac{\log|\Delta_N|}{n^2} \geq -\frac{1}{n_0^2} \int_0^{H_0} \mathcal{D} - \Big(\frac{\delta}{2} + \frac{\delta^2}{8}\Big)\log 2 - o(1).$$ □

3.4 Nonvanishing along the primes

Proposition 11 ([1, Prop. 4.1])**.** *Assume (C1), (C3) and $G \in \mathbb{Q}$. For every sufficiently large prime p, $v_p(\Delta_p) = -2n_0p$; in particular $\Delta_p \neq 0$.*

Proof. We go through [1, Sec. 4] with $N = p$ and the rows P_r, D_r of the design with $N = 1$, $0 \le r \le n_0$.

Bases and Frobenius. The palindromic bases and [1, Eqs. (49)–(51)] do not involve the design. Writing $r = pr_0 + i$ with $0 \le r_0 < n_0$, $0 \le i < p$, and using $C = C_0p$, $h = h_0p$, $g = g_0p$, [1, Eq. (52)] becomes $tR_rE = (1-t')^{h_0}(t')^{C_0}(w')^{r_0-g_0}E\,w^i$, and [1, Eqs. (53)–(55)] follow as in the preprint. The successor index $r_0 + 1 \le n_0$ is at most the auxiliary row.

Degrees. The auxiliary row has $\min\deg P_{n_0} = C_0 - 1 - (n_0 - g_0) = A_0 - 1$ and $\min\deg D_{n_0} = A_0$, the analogue of [1, Eq. (56)], and contact order $C_0 + n_0 - g_0 \ge L_0$. In the last block $r = (n_0-1)p+i$, both the successor contribution and the original rows start in degree $(A_0+1)p - i$, so the consistency check after [1, Eq. (56)] holds.

Range of primes. [1, Sec. 4.3] needs $p > 2\sqrt{H_0p}$, so that [1, Eqs. (31), (36)] apply at $N = p$, and $p > H_0$, so that the base entries, whose denominators have prime factors at most H_0, reduce modulo p. Both hold for $p > 4H_0$; the preprint's $p > 260$ is this condition for $H_0 = 65$.

Filter and blocks. In $\mathbb{F}_p[s]$, $s^{b_0p+k}(1-s)^{q_0p} = s^{\ell}(s^p)^{b_0+k_0}(1-s^p)^{q_0}$ for $k = \ell + pk_0$, which is [1, Eq. (58)]; it uses the base raw indices $b_0, \dots, L_0 - 1$. The reduction of the scaled matrix is $L_p = a^{\mathsf{T}} \otimes B_0 + b^{\mathsf{T}} \otimes B_1$ with B_0, B_1 of Definition 4, and [1, Eq. (60)] gives

$$\det L_p = (\det a)^{n_0} \det B_0 \, \det(B_0 + B_1)^{(p-1)/2} \det(B_0 - B_1)^{(p-1)/2}.$$

Conclusion. Let E be the finite set of primes consisting of 2, the primes dividing the denominator of G, the denominators of the entries of B, and the numerators and denominators of the three nonzero determinants of (C3). For $p > 4H_0$ outside E, $\det L_p \neq 0$, and the last paragraph of the proof of [1, Prop. 4.1] gives $v_p(\Delta_p) = -2n = -2n_0p$. □

3.5 The real place

Proposition 12 ([1, Props. 5.2, 5.3, 6.1])**.** *Assume (C1). For $\kappa \in \{1,2\}$ let $p^{(\kappa)}, v^{(\kappa)}$ be admissible and $\lambda^{(\kappa)} \in \Lambda_\kappa$. Then*

$$\limsup_{N\to\infty} \Big(\frac{\log|\Delta_N|}{n^2} - \frac{1}{2}\log 2 \Big) \le \max_{\kappa\in\{1,2\}} U_\kappa\big(p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}\big).$$

Proof. Section 5 of the preprint. Put $D_* = n - 1 - 2g$. By (C1), $2g \le n-2$, so $0 < D_* < n$, which is what [1, Eq. (61)] requires. The identity [1, Eq. (66)], $\xi_f x^{g-r} + \xi_n x^{r-g} = \xi_f x^{g-(n-1)}(x^{n-1-r} + c\,x^{D_*}x^r)$, holds for every design; the values $c \in \{1, -5\}$ come from the factor $3/2$ in the entries. [1, Lemma 5.1] uses $D_* < n$ in the bound $|F'/F| \le \frac{4}{3}D_* + \frac{8}{3}n < 4n$, and [1, Props. 5.2, 5.3] use $D_* \ge 0$. The identity [1, Eq. (65)], the majorants $I_{\kappa,n}$ of [1, Eq. (74)] and the estimate [1, Eq. (75)] then hold with the parameters of the design.

Exponents. Substituting [1, Eq. (79)] into [1, Eq. (74)] and collecting powers gives, with $\langle\cdot\rangle$ the empirical averages of [1, Sec. 6], the identity [1, Eq. (80)] in the form

$$\begin{aligned}\frac{\log I_{\kappa,n}}{n^2} - \frac{1}{2}\log 2 = c_{\kappa,n}\log 2 + \langle W_\kappa(x)\rangle + \frac{\kappa+2}{2n}\langle\log(1+x^2)\rangle + \langle W_s(s)\rangle \\ + \frac{\kappa}{2}\langle\log|x-x'|\rangle_{\neq} + \frac{1}{2}\langle\log(1-xx')\rangle_{\neq} + \langle\log|s-s'|\rangle_{\neq} - \langle\log(1-2xs+x^2)\rangle,\end{aligned}$$

with $c_{\kappa,n} = C/n + (\kappa-2)/2 - (\kappa+1)/(2n)$, $W_\kappa(x) = (\alpha+2\gamma_g)\log|x| + 2\eta\log(1-x) - (\kappa/2+\alpha+\eta+\gamma_g)\log(1+x^2)$ and $W_s(s) = \beta\log s + \gamma_q\log(1-s)$. Indeed, the exponent of $|x_i|$ is $(C-1)+g-(n-1) = a+2g$ in both majorants; the exponent of $(1+x_i^2)$ is $-(C+h)+2$ for $\kappa=2$ and $-(C+h)+2+(n-1)/2$ for $\kappa=1$, and $C/n = 1+\alpha+\gamma_g$; the factor $(1-t_i)^h$ gives $2\eta\log(1-x)$; and the s-weights come from $s^b(1-s)^q$. This is the only place where the preprint's single letter γ must be split: γ_g appears in W_κ and in C/n, and γ_q in W_s.

Constraint. Condition [1, Eq. (67)], $\sum_i (1-x_i^2)/(1+x_i^2) \le D_*$, is equivalent to $\langle 2\gamma_g - 2x^2/(1+x^2)\rangle \le -1/n$, because $(1-x^2)/(1+x^2) = 1 - 2x^2/(1+x^2)$, $D_* = n-1-2g$ and $2g = 2\gamma_g n$. Its failure therefore gives $\langle D\rangle > -1/n$ with $D(x) = 2\gamma_g - 2x^2/(1+x^2)$, and adding $\lambda\langle D\rangle$ with $\lambda \ge 0$ raises the bound by at most λ/n in the case $\kappa = 1$.

Section 6 of the preprint. The damped kernels, the identities [1, Eqs. (86), (87)] and the completion of squares do not involve the design. Adding the constants of [1, Eqs. (86), (87)] to $c_{\kappa,n}\log 2$ gives $(C/n + (\kappa-2)/2 - \kappa/2 - 1)\log 2 + O(1/n) = (-1+\alpha+\gamma_g)\log 2 + O(1/n)$. In the last step of the proof of [1, Prop. 6.1] the coefficients 19/48, 1/12, 7/48, 1/12 are $\alpha+2\gamma_g$, 2η, β, γ_q; they are positive because the design consists of positive integers, and the weakened coefficients $2\eta - 1/(2n)$ and $\gamma_q - 1/n$ are at least η and $\gamma_q/2$ for all large n. This gives the uniform decay to $-\infty$ near $x=0$, $x=1$, $s=0$, $s=1$ and the compactness used there. The conclusion of [1, Prop. 6.1] is the stated bound with U_κ of Definition 5, and [1, Eq. (75)] transfers it to Δ_N. □

3.6 Conclusion

Proof of Theorem 6. Suppose $G \in \mathbb{Q}$. By Proposition 8, $\Delta_N \in \mathbb{Q}$ for every N. By Proposition 11, $\Delta_p \neq 0$ for every sufficiently large prime p. Along this sequence of primes, Proposition 10 gives $\liminf(\log|\Delta_p|/n^2 - \frac{1}{2}\log 2) \ge F_{\mathrm{low}}$, and Proposition 12 with the sequences of (C4) gives $\limsup(\log|\Delta_p|/n^2 - \frac{1}{2}\log 2) \le \max_\kappa U_\kappa < F_{\mathrm{low}}$. This is a contradiction. □

4 The design $(96, 15, 6, 8, 5)$

In this section $(n_0, b_0, q_0, g_0, h_0) = (96, 15, 6, 8, 5)$. The derived integers are $a_0 = 21$, $L_0 = 117$, $C_0 = 125$, $H_0 = 130$, $A_0 = 37$, and the ratios are $\alpha = 21/96$, $\beta = 15/96$, $\gamma_q = 6/96$, $\gamma_g = 8/96$, $\eta = 5/96$, $\delta = 19/96$.

4.1 Conditions (C1) and (C2)

(C1) reads $16 < 95$ and (C2) reads $30 < 130$.

4.2 The finite-place constant

Proposition 13. *For the design* $(96, 15, 6, 8, 5)$ *the loss function* $\mathcal{D}$ *is continuous and is given by Table 1;* $\int_0^{130}\mathcal{D} = 17114$*, and*

$$F_{\mathrm{low}} = -\frac{8557}{4608} - \frac{44521}{73728}\log 2 = -2.27554803637\ldots$$

Proof. On each interval of Table 1 every min, max and positive part in Definition 3 is attained by a fixed one of its arguments, so $\mathcal{D}$ is linear there; the listed formulas and end values follow by direct substitution, and adjacent end values agree. The candidate breakpoints are the crossings of the linear forms appearing in Definition 3; the script `cat_cert7_new.py` recomputes the integral exactly over all of them, as does `cat_design_lower.py`, and both reproduce 8609/2 for the preprint's design. The constant follows from $\delta = 19/96$: $\frac{1}{2} + \frac{\delta}{2} + \frac{\delta^2}{8} = \frac{44521}{73728}$ and $17114/96^2 = 8557/4608$. □

x range	$\mathcal{D}(x)$	end values	$\int$
[0, 99/2]	192	192, 192	9504
[99/2, 115/2]	$291 - 2x$	192, 176	1472
[115/2, 65]	176	176, 176	1320
[65, 68]	$306 - 2x$	176, 170	519
[68, 80]	$442 - 4x$	170, 122	1752
[80, 115]	$362 - 3x$	122, 17	4865/2
[115, 117]	$247 - 2x$	17, 13	30
[117, 130]	$130 - x$	13, 0	169/2
total			17114

Table 1: The loss function of the design $(96, 15, 6, 8, 5)$, the analogue of [1, Table (43)]. The slopes $0, -2, 0, -2, -4, -3, -2, -1$ are those of the preprint's table.

4.3 Condition (C3)

Proposition 14. *For the design* $(96, 15, 6, 8, 5)$ *the three matrices* B_0, B_0+B_1, B_0-B_1 *of Definition 4 are invertible over* $\mathbb{Q}$.

Proof. The entries of B are built by the rational recipe of [1, Sec. 4.4] with the arrays extended to $0 \le i \le H_0 - 1 = 129$, $0 \le j \le L_0 - 1 = 116$, the Chebyshev recursions run to $m = n_0 - g_0 = 88$, and $q_0 = 6$ successive differences. Every scalar denominator in the recipe is a product of nonzero integers of absolute value at most $H_0 = 130$, so every denominator of B is a unit modulo the prime 131, and the construction can be carried out over $\mathbb{F}_{131}$. Gaussian elimination by the rule of [1, Sec. 4.4] produces the nonzero pivots listed in Table 3 (Appendix A). Hence the three matrices are invertible modulo 131, and their rational determinants are nonzero. □

The script `cat_cert4_new.py` builds B in exact rational arithmetic, checks the rows against [1, Eq. (3)] and the vanishing of the $\zeta(2)$ coefficient in all 97×102 raw entries, finds the largest denominator prime of B to be 127, and computes the pivots. Run on the preprint's design modulo 101, the same code reproduces all 144 pivots and both row swaps of [1, Table 1].

4.4 Condition (C4)

The trial sequences have the form used in [1, Sec. 7]: for $u \in \{p, v\}$,

$$u_k = l_k\,[k \le 16] + \sum_{z \in Z_{\mathbb{R}}} r_z z^k + \sum_{z \in Z_{\mathbb{C}}} \left(a_z \operatorname{Re} z^k + b_z \operatorname{Im} z^k\right) \qquad (k \ge 1),$$

with the 18 real bases

$$\begin{aligned} Z_{\mathbb{R}} = \{ &-0.7, -0.85, -0.93, -0.97, -0.99, -0.997, -0.999, \\ &0.5, 0.8, 0.9, 0.95, 0.98, 0.99, 0.993, 0.995, 0.998, 0.999, 0.9995\} \end{aligned}$$

and the 19 complex bases $Z_{\mathbb{C}} = \{z_1, \ldots, z_{19}\}$ of Table 2. All bases have modulus less than 1, so the sequences are admissible. The 72 coefficients of each sequence and the multiplier λ for $\kappa = 1$ are the exact rational numbers of Tables 4 and 5 (Appendix B).

The bases are placed at the support edges of the equilibrium measures of Section 6. The optimal sequences are moments of those measures (Proposition 45); an edge at $x = \cos\varphi$ contributes to the

Chebyshev moments terms of the form $k^{-\sigma}\cos(k\varphi + c)$, and $k^{-\sigma}$ is a superposition of geometric sequences e^{-kt}. Each edge angle therefore receives a few radii approaching 1: the negative real bases for the edge at $x = -1$, the complex bases $z_1, \dots, z_{10}$ for the two edges of the gap around $x = 0$, $z_{11}, \dots, z_{14}$ for the upper edge of μ near 0.993, $z_{15}, \dots, z_{19}$ for the lower edge of ν near 0.0065, and the positive real bases for the edges near 1. This choice is a heuristic; the proof uses only the resulting rational numbers.

	base	modulus	argument
z_1	$0.0834 + 0.8459i$	0.850	84.37°
z_2	$0.0912 + 0.9255i$	0.930	84.37°
z_3	$0.0952 + 0.9653i$	0.970	84.37°
z_4	$0.0971 + 0.9852i$	0.990	84.37°
z_5	$0.0978 + 0.9922i$	0.997	84.37°
z_6	$-0.0859 + 0.8456i$	0.850	95.80°
z_7	$-0.0940 + 0.9252i$	0.930	95.80°
z_8	$-0.0981 + 0.9650i$	0.970	95.80°
z_9	$-0.1001 + 0.9849i$	0.990	95.80°
z_{10}	$-0.1008 + 0.9919i$	0.997	95.80°
z_{11}	$0.9435 + 0.1106i$	0.950	6.69°
z_{12}	$0.9733 + 0.1141i$	0.980	6.69°
z_{13}	$0.9833 + 0.1153i$	0.990	6.69°
z_{14}	$0.9902 + 0.1161i$	0.997	6.69°
z_{15}	$0.0055 + 0.8500i$	0.850	89.63°
z_{16}	$0.0060 + 0.9300i$	0.930	89.63°
z_{17}	$0.0063 + 0.9700i$	0.970	89.63°
z_{18}	$0.0064 + 0.9900i$	0.990	89.63°
z_{19}	$0.0065 + 0.9970i$	0.997	89.63°

Table 2: Complex bases of the trial sequences (exact decimals). $z_1, \dots, z_5$ lie at the angle $\arccos 0.0981$, $z_6, \dots, z_{10}$ at $\arccos(-0.1011)$, $z_{11}, \dots, z_{14}$ at $\arccos 0.9932$, $z_{15}, \dots, z_{19}$ at $\arccos 0.0065$, the support edges found in Section 6.

In closed form,

$$T(u,x) = \sum_{k\le 16} \frac{l_k T_k(x)}{k} - \frac{1}{2}\sum_z \operatorname{Re}\left(\rho_z \operatorname{Log}(1 - 2xz + z^2)\right),$$

$$S(u,x) = \sum_{k\le 16} \frac{l_k x^k}{k} - \sum_z \operatorname{Re}\left(\rho_z \operatorname{Log}(1 - xz)\right),$$

$$\|u\|_*^2 = \sum_{k\le 16} \frac{l_k^2 + 2l_k t_k}{k} - \sum_{z,z'} \rho_z \rho_{z'} \operatorname{Log}(1 - zz'), \qquad t_k = \sum_z \rho_z z^k,$$

where the sums over z run over $Z_{\mathbb{R}}$ with $\rho_z = r_z$ and over $Z_{\mathbb{C}}$ and its conjugates with $\rho_z = (a_z - ib_z)/2$, $\rho_{\bar z} = \overline{\rho_z}$. Each factor $1 - ze^{\pm i\theta}$ with $|z| < 1$ has positive real part, so the principal logarithm of $1 - 2xz + z^2 = (1 - ze^{i\theta})(1 - ze^{-i\theta})$, $x = \cos\theta$, is the sum of the two principal logarithms, and the formulas follow from [1, Eq. (82)].

Proposition 15. *With the sequences of Tables 4 and 5,*

$$U_2 \le -2.27813351541685, \qquad U_1 \le -2.41461192963326.$$

Consequently $\max(U_1, U_2) < F_{\text{low}}$, *with margin* $F_{\text{low}} - \max(U_1, U_2) > 0.0025854$.

Proof. The proof is the computation performed by `cat_cert7_new.py`, which follows [1, Secs. 7.2–7.4] and replaces the rational logarithm recipe of [1, Sec. 7.3] by interval arithmetic. For each κ:

1. *Derivative numerators.* $X' = A_X/Q_X$ and $Y' = A_Y/Q_Y$ with polynomials over $\mathbb{Q}(i)$ computed exactly: $Q_X = x(1-x)(1+x^2)^{3-\kappa}\prod(1-2xz+z^2)\prod(1-xz)$, the products over the bases of p and v respectively with conjugates, and $Q_Y = s(1-s)\prod(1-2sz+z^2)$ over the bases of v. All divisions are exact and the imaginary parts cancel. The degrees of (A_X, Q_X, A_Y, Q_Y) are $(131, 116, 73, 58)$ for $\kappa = 2$ and $(133, 118, 73, 58)$ for $\kappa = 1$. Every factor of Q_X and Q_Y other than $x(1-x)$, $s(1-s)$ is positive on $[-1, 1]$, because $|z| < 1$, so X and Y are smooth on $(-1, 0)$, $(0, 1)$.

2. *Stationary points.* All real roots of A_X in $(-1, 0)$ and $(0, 1)$ and of A_Y in $(0, 1)$ are isolated by Descartes' rule of signs with bisection (the Vincent–Collins–Akritas method) in exact integer arithmetic, and each is refined to a bracket of width at most 2^{-50} with a strict sign change of the numerator at its ends. The bisection terminates with counts 0 or 1 on every piece, so all roots are simple and every stationary point is bracketed. There are 26 stationary points of X and 39 of Y for $\kappa = 2$, and 32 and 37 for $\kappa = 1$.

3. *Global suprema.* Since $\alpha+2\gamma_g$, η, β, γ_q are positive and the other terms are bounded, $X \to -\infty$ at $x \to 0$ and $x \to 1$, and $Y \to -\infty$ at $s \to 0$ and $s \to 1$. Hence $\sup X$ is the larger of $X(-1)$ and the values at the stationary points, and $\sup Y$ is the largest value at a stationary point.

4. *Values.* On a bracket $[\ell, \ell + w]$, $X \le X(\ell) + w \sup|X'|$. Here $X(\ell)$ is enclosed in interval arithmetic at 60 significant digits (`mpmath.iv`), and X' is enclosed on the whole bracket by interval evaluation of its closed form, which is checked against A_X/Q_X at five points to relative error below 10^{-35}. The largest allowance $w \sup|X'|$ is 3.7×10^{-22}. The norms and $\log 2$ are enclosed in the same arithmetic.

The upper endpoints give

	$\sup X$	$\sup Y$	$\kappa\|p\|_*^2 + \frac{1}{2}\|v\|_*^2$
$\kappa = 2$	≤ -0.97899971671247	≤ -1.59600372010315	≤ 0.78062889116456
$\kappa = 1$	≤ -1.40122170303065	≤ -1.58246983871889	≤ 1.05283858188207

Adding $(-1+\alpha+\gamma_g)\log 2 = -\frac{67}{96}\log 2$ gives the stated bounds. The finite-place constant is enclosed in the same arithmetic as $F_{\text{low}} \ge -2.27554803637302$. □

As controls, the script compares A/Q with numerical derivatives of X and Y (relative error below 10^{-40}) and compares each certified supremum with the maximum over a grid of 200,000 points in floating point (agreement within 4×10^{-10}). The coefficients were produced by `cat_tail_edges.py`, which minimises the bound over the coefficients for the fixed bases, a convex problem, and rounds the result; that computation is not part of the proof.

Remark 16 (A certificate with fewer coefficients)**.** With 46 coefficients per sequence (finite part of length 16, the 12 real bases ± 0.5, ± 0.8, ± 0.9, ± 0.95, ± 0.98, 0.99, 0.995 and the 9 complex bases $0.6i$, $0.8i$, $0.9i$, $0.95i$, $0.98i$, $\pm 0.1 + 0.9i$, $\pm 0.1 + 0.95i$, not tied to the support edges), the same certificate gives $U_2 \le -2.27779256156814$, $U_1 \le -2.41413175743286$ and margin 0.0022445. Its coefficients are in `cat_tail_opt_out.json` (SHA-256 beginning `5841a0f1`); `cat_cert7_new.py` certifies both files in one run.

Theorem 17. *The design* $(96, 15, 6, 8, 5)$ *satisfies (C1)–(C4). Hence, by Theorem 6, Catalan's constant is irrational, with the gap between the two constants exceeding* 0.0025854.

Proof. Combine Section 4.1 and Propositions 13, 14 and 15. □

4.5 Design search

Observation 18 (Numerical; not part of the proof)**.** For each design we computed F_{low} exactly and minimised U_2 numerically over the first 300 coefficients of p and v, scoring the design by $F_{\mathrm{low}} - U_2$. A local search over designs (`cat_design_search.py`, log of 7 October 2026) starting from $(48, 7, 4, 4, 2)$, whose score is 0.00111, reached $(48, 8, 3, 4, 3)$ with score 0.00225 at size 48 and $(96, 15, 6, 8, 5)$ with score 0.00244 at size 96. For $(96, 15, 6, 8, 5)$ the script `cat_newdesign_check.py` also computed Δ_1 exactly with the hypothetical value $G = 1/3$ and found, for each of the 23 primes $2\sqrt{130} < p \le 130$, $-v_p(\Delta_1) \le \mathcal{D}(p)$, with equality at 11 of them.

5 An irrationality exponent

The comparison of Theorem 6 also bounds rational approximations to G. Jha [9] obtains $\mu(G) \le 10^8$ for the design $(48, 7, 4, 4, 2)$ from three ingredients: denominator bounds that hold at every rational argument, nonvanishing at prime scales that avoid the denominator, and bounds for all complementary minors of the determinant. In this section we check these ingredients for a design satisfying (C1)–(C4), bound the minors of every size by an energy estimate for configurations with fewer nodes that uses the certificate of (C4), sharpen the bounds of [9] for the row and column determinants (Section 5.5), use the bound of [10] for the degree of the determinant as a polynomial in the constant (Lemma 20), and obtain, for the design $(96, 15, 6, 8, 5)$, an exponent below 2 715 (Theorem 35). Litsey [10] obtains $\mu(G) \le 8\,127$ for the design $(48, 7, 4, 4, 2)$ by a different perturbation estimate: he writes $\Phi^{(1)}$, by a quadrature formula, through the values of the rows at fixed nodes, expands the determinant column by column, and bounds the resulting determinants, whose rows keep their full number, by the energy of fewer points; he also certifies a gap of 1.0×10^{-3} for that design. Section 5.7 records measurements of the determinants near G: they indicate that the method gives about 2.1×10^3, and they show which bounds account for the difference. Throughout, the design satisfies (C1)–(C4), we fix the sequences of (C4), and we put

$$U = \max_{\kappa \in \{1,2\}} U_\kappa(p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}),$$

$$c_{\mathrm{low}} = F_{\mathrm{low}} + \tfrac{1}{2}\log 2, \qquad c_{\mathrm{up}} = U + \tfrac{1}{2}\log 2, \qquad \gamma = c_{\mathrm{low}} - c_{\mathrm{up}} > 0.$$

For the design $(96, 15, 6, 8, 5)$, $F_{\mathrm{low}} = -2.27554803637\ldots$ and $\gamma > 0.0025854$ (Theorem 17).

The argument has four steps. First, arithmetic: at every rational argument a/b and at prime scales $p \nmid b$, the value $\Delta_p(a/b)$ is nonzero and its size is bounded below through the denominators of its entries (Sections 5.1 and 5.2). Second, size: the entries of $\Phi^{(1)}$ are bounded (Section 5.3), and every minor of $A = \Phi^{(0)} + G\Phi^{(1)}$ is bounded through an energy of fewer nodes (Section 5.4) and bounds for the row and column determinants (Section 5.5). Third, perturbation: these bounds give an upper bound for $|\Delta_p(x)|$ at every x within distance $e^{-\rho n}$ of G. Fourth, comparison: a rational number that close to G contradicts the lower bound, which gives the exponent $\rho(1 - \gamma_g + n_0\gamma)/\gamma$ (Section 5.6). Section 5.7 compares the proved rate ρ with measurements.

5.1 The determinant as a polynomial

Let c_l be as in Section 2: $c_l = \binom{2l}{l}4^{-l}$ for integers $l \ge 0$, so that $0 < c_l \le 1$, and $c_l = 0$ otherwise.

Definition 19 (Determinant polynomial)**.** For $0 \le r < n$ and $b \le j < L$ let $F_j(r) = M(P_r, s^j) - \frac{3}{2} Z(D_r, s^j)$ be the raw entry [1, Sec. 2], with M, Z and the rows of Definition 2. By [1, Prop. 2.1] and Proposition 8 it has the form

$$F_j(r) = F_j^{(0)}(r) + G\, F_j^{(1)}(r),$$
$$F_j^{(0)}(r) = M^0(P_r, j) - \tfrac{3}{2} Z^0(D_r, j), \qquad F_j^{(1)}(r) = 4 \sum_v [t^v] P_r \; c_{(v-j)/2},$$

with $c_{(v-j)/2} = 0$ unless $v - j$ is even and nonnegative [1, Sec. 4.3]. For $e \in \{0, 1\}$ and $0 \le r, k < n$ let

$$\Phi_{r,k}^{(e)} = \sum_{u=0}^{q} (-1)^u \binom{q}{u} F_{b+k+u}^{(e)}(r),$$

and let $\Delta_N(X) = \det\left(\Phi^{(0)} + X\Phi^{(1)}\right) \in \mathbb{Q}[X]$. Then $\Delta_N(G) = \Delta_N$ is the determinant of [1, Eq. (7)].

Lemma 20 (Degree; [10, Lemma 3.1])**.** *$\Delta_N(X)$ has degree at most $n - g$.*

Proof. By Definition 2, $P_r = (1-t)^h t^{C-1} T_{|r-g|}(1/t)$ depends on r only through $|r - g|$, and $F_j^{(1)}(r)$ depends on r only through P_r; so rows r and $2g - r$ of $\Phi^{(1)}$ are equal for $0 \le r < g$, and $2g - r < n$ by (C1). Subtracting row r from row $2g - r$ for $0 \le r < g$ does not change $\Delta_N(X)$ and makes the g rows of index $g + 1, \dots, 2g$ independent of X. Expanding the determinant along its rows gives a polynomial of degree at most $n - g$. □

5.2 Arithmetic at integer and rational arguments

Lemma 21 (Local bounds at integers)**.** *There are integers $D_{N,p} \ge 0$, defined for every positive integer N and every prime p, with $D_{N,p} = 0$ for $p > H$, such that*

1. *$v_p(\Delta_N(x)) \ge -D_{N,p}$ for every integer x and every prime p;*
2. *$\sum_p D_{N,p} \log p \le -c_{\mathrm{low}} n^2 + o(n^2)$ as $N \to \infty$.*

Proof. We go through the proofs of Propositions 9 and 10, which follow [1, Secs. 2.2 and 3], with G replaced by an integer x. In those proofs the constant enters in three ways only. (i) At odd primes $2\sqrt{H} < p \le H$ the term $G\, F_j^{(1)}(r)$ is used only through the integrality of $F_j^{(1)}(r)$ at p, which holds because every c_l is p-integral at odd p; after multiplication by p or p^2 this term vanishes modulo p [1, Sec. 3.2, Lemmas 3.2, 3.3]. The same holds with G replaced by $x \in \mathbb{Z}$. (ii) At odd primes $p \le 2\sqrt{H}$ the entry bound of [1, Sec. 3.4] reads $v_p(F_j(r)) \ge -C_0(1 + \log H / \log p) - v_p(\operatorname{den} G)$; for $x \in \mathbb{Z}$ the last term is zero. At primes $p > H$ every entry is p-integral, because the factorials and degree denominators have prime factors at most H and x has none. (iii) At $p = 2$ the constant enters only through $e_1 = 4G - M_s(0,0)$ and $e_2 = -M_s(0,1)$ of [1, Eq. (24)], through a fixed integer K_G with $v_2(e_1), v_2(e_2) \ge -K_G$. For $x \in \mathbb{Z}$, $v_2(4x - M_s(0,0)) \ge \min\left(2, v_2(M_s(0,0))\right)$, so one integer K serves for all x, and the bound of Proposition 9 holds for $\Delta_N(x)$ with an error term $O(n \log(n+2))$ independent of x. Let $D_{N,p}$ be the negatives of the resulting lower bounds, rounded up to integers and replaced by 0 when negative. They do not depend on x, which is (1). The sum in (2) is the sum estimated in the proof of Proposition 10, whose value is $-c_{\mathrm{low}} n^2 + o(n^2)$. □

Lemma 22 (Rational arguments)**.** *Let $P \in \mathbb{Q}[X]$ have degree at most d, let p be a prime and $D \ge 0$ an integer with $v_p(P(x)) \ge -D$ for every integer x. Then $v_p\big(b^d\, d!\, P(a/b)\big) \ge -D$ for all integers a and $b \ge 1$.*

Proof. Write $P = \sum_{k=0}^{d} e_k \binom{X}{k}$ with $e_k = \sum_{i=0}^{k} (-1)^{k-i}\binom{k}{i} P(i)$. Then $v_p(e_k) \geq -D$. For $0 \leq k \leq d$,

$$b^d\, d! \binom{a/b}{k} = b^{d-k}\, \frac{d!}{k!} \prod_{i=0}^{k-1} (a - ib) \in \mathbb{Z}.$$

Hence $b^d d! P(a/b) = \sum_k e_k \cdot (\text{integer})$ has $v_p \geq -D$. □

Corollary 23 (Lower bound at rational arguments)**.** *For all integers a and $b \geq 1$ with $\Delta_N(a/b) \neq 0$,*

$$\log |\Delta_N(a/b)| \geq c_{\text{low}} n^2 - (n-g)\log b - o(n^2),$$

where the term $o(n^2)$ does not depend on a and b.

Proof. By Lemmas 20, 21 and 22, with $d = n - g$, $b^d\, d! \prod_p p^{D_{N,p}}\, \Delta_N(a/b)$ is a nonzero integer. Its absolute value is at least 1; take logarithms and use $\log d! = O(n \log n)$. □

Lemma 24 (Nonvanishing)**.** *There is a finite set E_0 of primes, depending only on the design, such that for every prime $p > 4H_0$ with $p \notin E_0$ and all integers a and $b \geq 1$ with $p \nmid b$, $v_p(\Delta_p(a/b)) = -2n_0 p$. In particular $\Delta_p(a/b) \neq 0$.*

Proof. Let E_0 consist of 2, the primes dividing the denominators of the entries of the fixed matrix B of Definition 4, and the primes dividing the numerators and denominators of the three determinants of (C3). The proof of Proposition 11, which follows [1, Sec. 4], uses the constant only in [1, Sec. 4.3]: there the term $G\, F_j^{(1)}(r)$ is p-integral and vanishes after multiplication by p^2, and the primes dividing the denominator of G are excluded. With G replaced by a/b and $p \nmid b$, the term $(a/b)F_j^{(1)}(r)$ is p-integral, and the proof gives $v_p(\Delta_p(a/b)) = -2n = -2n_0 p$. □

5.3 Entries

Lemma 25 (Entries)**.** *Put $d_r = |r - g|$. There is a constant c_1 depending only on the design such that for all N and $0 \leq r, k < n$,*

$$|\Phi_{r,k}^{(0)} + G\, \Phi_{r,k}^{(1)}| \leq c_1\, 2^h (1+\sqrt{2})^{d_r}, \qquad |\Phi_{r,k}^{(1)}| \leq c_1\, 2^{h+q} (1+\sqrt{2})^{d_r}.$$

Proof. Let $\tilde{T}_d(x) = i^{-d} T_d(ix)$ and $\tilde{U}_d(x) = i^{-d} U_d(ix)$. They satisfy $\tilde{T}_{d+1} = 2x\tilde{T}_d + \tilde{T}_{d-1}$ and the same recurrence for $\tilde{U}$, with $\tilde{T}_0 = 1$, $\tilde{T}_1 = x$, $\tilde{U}_0 = 1$, $\tilde{U}_1 = 2x$, so their coefficients are nonnegative and equal to the absolute values of those of T_d and U_d. Hence the sums of the absolute values of the coefficients are $\tilde{T}_d(1)$ and $\tilde{U}_{d-1}(1)$, which satisfy $u_{d+1} = 2u_d + u_{d-1}$; since $(1+\sqrt{2})^2 = 2(1+\sqrt{2}) + 1$, induction from the initial values gives $\tilde{T}_d(1) \leq (1+\sqrt{2})^d$ and $\tilde{U}_{d-1}(1) \leq (1+\sqrt{2})^d$. By Lemma 7, $P_r = (1-t)^h t^{C-1} T_{d_r}(1/t)$ and $D_r = \pm(1-t)^h t^{C-1} U_{d_r-1}(1/t)$, so the sums $\|P_r\|_1$, $\|D_r\|_1$ of the absolute values of their coefficients are at most $2^h(1+\sqrt{2})^{d_r}$, and $|P_r(t)| \leq \|P_r\|_1$, $|D_r(t)| \leq \|D_r\|_1$ for $|t| \leq 1$.

By [1, Sec. 5.1], $\Phi_{r,k}^{(0)} + G\,\Phi_{r,k}^{(1)} = \int_{-1}^{1} \int_0^1 A_r(t)\, s^{b+k}(1-s)^q (1-ts)^{-1}\, ds\, d\mu(t)$ with $A_r(t) = P_r(t) - \mathbf{1}_{t>0} \frac{3}{2} f(t) D_r(t)/t$, $f(t) = \sqrt{1-t^2}$ and $d\mu(t) = |t|\, dt/f(t)$. The part with P_r is at most $\|P_r\|_1 K_\mu$, where $K_\mu = \iint (1 - |t|s)^{-1} ds\, d\mu(t) < \infty$ [1, Sec. 5.1]. The part with D_r is at most $\frac{3}{2}\|D_r\|_1 \int_0^1 \int_0^1 (1-ts)^{-1} ds\, dt = \frac{3}{2}\zeta(2)\|D_r\|_1$. This gives the first bound with $c_1 = K_\mu + \frac{3}{2}\zeta(2) + 4$. For the second, $|F_j^{(1)}(r)| \leq 4\|P_r\|_1$ because $0 \leq c_l \leq 1$, and the filter has $\sum_u \binom{q}{u} = 2^q$. □

5.4 Energy of minors

Jha [9, Sections 5–6] bounds every complementary minor of the matrix $A = \Phi^{(0)} + G\Phi^{(1)}$, whose determinant is Δ_N, through an energy of m row nodes and m column nodes. Its starting point is the integral representation of Section 2.

Lemma 26 (Selected minors; [9, Lemma 6.1])**.** *For injective lists $I = (r_1, \dots, r_m)$ and $J = (c_1, \dots, c_m)$ of indices in $\{0, \dots, n-1\}$,*

$$\det A[I,J] = \frac{1}{(m!)^2} \int_{(-1,1)^m} \int_{(0,1)^m} \det[A_{r_a}(t_i)]_{a,i} \, \frac{V(t)V(s)}{\prod_{i,j}(1 - t_i s_j)} \, \det[\psi_{c_b}(s_j)]_{j,b} \, ds \, d\mu^m(t),$$

with $V(t) = \prod_{i<j}(t_j - t_i)$, and the integrand is absolutely integrable.

Proof. The entries of $A[I,J]$ are $\iint A_{r_a}(t)\psi_{c_b}(s)(1-ts)^{-1} ds \, d\mu(t)$ (Section 2). The twofold Andréief identity in dimension m gives the factor $(m!)^{-2}$ and the determinant $\det[(1 - t_i s_j)^{-1}]$, which equals $V(t)V(s)/\prod_{i,j}(1 - t_i s_j)$ by Cauchy's double alternant [1, Eq. (63)]. Absolute integrability follows from the finiteness of $\iint (1 - |t|s)^{-1} ds \, d\mu(t)$, as in [1, Sec. 5.1]. □

In the coordinate $t = t(x) = 2x/(1+x^2)$, $x \in (-1,1)$, one has $d\mu(t(x)) = 4|x|(1+x^2)^{-2}dx$ and $A_r(t(x)) = (1-t)^h t^{C-1}(\xi_f(x)x^{g-r} + \xi_n(x)x^{r-g})$ with $(\xi_f, \xi_n) = (\frac{1}{2}, \frac{1}{2})$ for $x < 0$ and $(-\frac{1}{4}, \frac{5}{4})$ for $x > 0$ [1, Sec. 5.1]. For nodes $x_1, \dots, x_m \in (-1,1) \setminus \{0\}$ and $s_1, \dots, s_m \in (0,1)$ put, with $V(y) = \prod_{i<j} |y_j - y_i|$ for a real list y,

$$J_m(x,s) = \frac{V(t(x))\, V(s)^2}{\prod_{i,j}(1 - t(x_i)s_j)} \prod_i |t(x_i)|^{C-1}(1 - t(x_i))^h \prod_j s_j^b (1-s_j)^q,$$

and define the energies [9, Eqs. (5.1)–(5.3)]

$$E_{2,m}(x,s) = J_m(x,s)\, V(x) \prod_i |x_i|^{g-(n-1)},$$
$$E_{1,m}(x,s) = J_m(x,s)\, 2^{-\binom{m}{2}} \prod_i (1+x_i^2)^{(n-1)/2} |x_i|^{g-(n-1)},$$

with the ambient n, not m, in the one-node exponents; both equal 1 for $m = 0$. After the bounds for the row and column determinants of Section 5.5, the absolute integrand of Lemma 26 is at most a factor depending on m and n times $E_{2,m}$ when $\sum_i (1-x_i^2)/(1+x_i^2) \le D_* = n - 1 - 2g$, and times $E_{1,m}$ otherwise (Corollary 31). Jha writes $E_{\kappa,m}$ as a product of one-node factors $u_\kappa(x_i)$, $w(s_j)$ and pair factors [9, Eq. (5.8)], and for $m < n$ [9, Proposition 5.4] adjoins nodes until the size is n and applies the bound for $m = n$. We bound $E_{\kappa,m}$ directly, with the certificate of (C4).

Proposition 27 (Energy of fewer nodes)**.** *Let $p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}$ be the sequences of (C4), and put, for $\kappa \in \{1,2\}$,*

$$\Sigma_\kappa = \sup_x X_{p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}}(x) + \sup_s Y_{v^{(\kappa)}}(s),$$
$$\overline{U}_\kappa(\theta) = \theta\big((1 + \alpha + \gamma_g)\log 2 + \Sigma_\kappa\big) - \tfrac{3}{2}\theta^2 \log 2 + \kappa \|p^{(\kappa)}\|_*^2 + \tfrac{1}{2}\|v^{(\kappa)}\|_*^2,$$

with X, Y and $\|\cdot\|_*$ as in Definition 5. For every $\eta > 0$ there is N_η such that for $N \ge N_\eta$, every $m \le n$ with $\theta = m/n$ and every configuration of m row nodes and m column nodes,

$$\frac{\log E_{2,m}(x,s)}{n^2} \le \overline{U}_2(\theta) + \eta, \qquad \frac{\log E_{1,m}(x,s)}{n^2} \le \overline{U}_1(\theta) - \lambda^{(1)}(1 - 2\gamma_g)(1-\theta) + \eta,$$

the second whenever $\sum_i (1-x_i^2)/(1+x_i^2) > D_*$. At $\theta = 1$, $\overline{U}_\kappa(1) = U_\kappa(p^{(\kappa)}, v^{(\kappa)}, \lambda^{(\kappa)}) + \frac{1}{2}\log 2$.

Proof. Write $\log E_{\kappa,m}$ as a sum of one-node terms over the m nodes of each list and of pair terms over the pairs. Per node, $\log u_2(x) = (C-1)\log 2 + A\log|x| + 2h\log(1-x) - (C-1+h)\log(1+x^2)$ and $\log w(s) = b\log s + q\log(1-s)$. The exponents of $|x_i|$ and $1+x_i^2$ are those of the ambient size n [9, Eq. (5.8)], so the exponent of $|x_i|$ is $A = a + 2g$, as in [1, Eq. (80)]. With $C/n = 1 + \alpha + \gamma_g$ the one-node factors contribute $n\big((1+\alpha+\gamma_g)\log 2 + W_2(x)\big) + O(1)$ and $nW_s(s)$, where the term $-\log(1+x^2)$ of each pair factor R_2 is cancelled by the term $+\log(1+x^2)$ of each K, as in [9, Eq. (5.8)] for $m = n$; there are $m-1$ of each per node. Hence the one-node part, divided by n^2, is $\theta\big((1+\alpha+\gamma_g)\log 2 + \langle W_2\rangle_\mu + \langle W_s\rangle_\nu\big) + O(1/n)$, with μ, ν the empirical probability measures of the two lists. The pair part, divided by n^2, is θ^2 times $\frac{1}{2}\log 2$, from the factor 2 in $t(x) - t(y) = 2(x-y)(1-xy)/((1+x^2)(1+y^2))$ for $\kappa = 2$ (for $\kappa = 1$ the factor $\frac{1}{2}$ in R_1 removes it and $\frac{\kappa}{2}$ changes accordingly), plus the off-diagonal averages of $\frac{\kappa}{2}\log|x-x'| + \frac{1}{2}\log(1-xx')$, $\log|s-s'|$ and $-\log(1-2xs+x^2)$.

The proof of [1, Prop. 6.1], which is Proposition 12 for a general design, expands these kernels in Chebyshev and power series with damping, bounds the omitted diagonals and the endpoint terms by $O(n\log n)$ uniformly in the configuration, and completes squares: with moments $a_k = \theta A_k$ of the measures of mass θ, it uses $-a^2 \le p^2 - 2pa$, so that $-\kappa\theta^2\sum_k A_k^2/k \le \kappa\|p\|_*^2 - 2\kappa\theta\sum_k p_k A_k/k$, and similarly for v. Each step holds for $m < n$. The averages of [1, Eq. (80)] are sums over the nodes divided by n (by n^2 for pairs), so with m nodes they are averages against measures of mass θ. The damped kernels of [1, Eqs. (83)–(85)] are compared pair by pair with constants independent of the configuration. The identities [1, Eqs. (86), (87)] are exact for any number of nodes. The m omitted diagonals cost $O_\varepsilon(1/n)$, and so do the terms $-\frac{1}{2n}\langle\log(1-x)\rangle$ and $-\frac{1}{n}\langle\log(1-s)\rangle$, as for $m = n$. A one-node average over m nodes is at most θ times the supremum of the function.

The constant terms of the expansions, $-\frac{\kappa}{2}\log 2$ from $\log|x-x'|$ and $-\log 2$ from $\log|s-s'|$, are multiplied by θ^2; with the pair constant they give $-\frac{3}{2}\theta^2\log 2$ for both κ. The remaining terms are θ times one-node averages bounded by Σ_κ. For $\kappa = 1$ the failure of the condition reads $\theta\langle 1 - 2x^2/(1+x^2)\rangle > 1 - 2\gamma_g - 1/n$, that is $\theta\langle D\rangle > (1-2\gamma_g)(1-\theta) - 1/n$ with D as in Section 6; adding $\lambda^{(1)}$ times the difference absorbs $\lambda^{(1)}\theta\langle D\rangle$ into X and leaves $-\lambda^{(1)}(1-2\gamma_g)(1-\theta) + O(1/n)$. At $\theta = 1$ the bound is $(-\frac{1}{2} + \alpha + \gamma_g)\log 2 + \kappa\|p\|_*^2 + \frac{1}{2}\|v\|_*^2 + \Sigma_\kappa = U_\kappa + \frac{1}{2}\log 2$. □

5.5 Alternants

The proof of [9, Theorem 6.3] bounds the row and column determinants with the losses $L_{n,m} = 2^{mk}(k+1)^n$ of [9, Lemma 4.1, Corollary 4.2] and 14^{nk} of [9, Lemma 4.5], where $k = n - m$. The lemmas of this subsection replace them by a factor $\max_{|z|=1}\prod_i |1 + x_i z|$ per removed index, which is at most 2^m for nodes in $(-1, 1)$. For real numbers $x_1, \dots, x_m$ put

$$V(x) = \prod_{i<j} |x_j - x_i|, \qquad \Theta(x) = \max_{|z|=1} \prod_{i=1}^{m} |1 + x_i z|.$$

Since $\prod_i |1 - x_i z| = \prod_i |1 + x_i(-z)|$, $\Theta(x)$ is also the maximum of $\prod_i |1 - x_i z|$ on the unit circle. For $x_i \in (-1, 1)$, $\Theta(x) \le 2^m$; for $x_i \in (0, 1)$, $\Theta(x) = \prod_i (1 + x_i)$, the maximum being attained at $z = 1$.

Lemma 28 (Monomial alternants)**.** *Let $0 \le m \le n$, $k = n - m$, let $r_1, \dots, r_m$ be distinct elements of $\{0, \dots, n-1\}$ and let $x_1, \dots, x_m$ be real. Then*

$$\big|\det[x_i^{r_j}]_{i,j=1}^m\big| \le V(x)\,\Theta(x)^k.$$

Proof. If two nodes coincide both sides vanish, and for $k = 0$ the determinant is $\pm V(x)$. Otherwise order the exponents as $r_{(1)} > \cdots > r_{(m)}$ and put $\lambda_j = r_{(j)} - (m - j)$. Then λ is a partition with at

most m parts and $\lambda_1 \le n-1-(m-1) = k$, and by the bialternant formula $\det[x_i^{r_j}] = \pm V(x)\, s_\lambda(x)$, with s_λ the Schur polynomial [11, Chapter I, Section 3]. The conjugate partition λ' has at most k parts. By the dual Cauchy identity [11, Chapter I, Section 4], for $y \in \mathbb{C}^k$,

$$\prod_{i=1}^{m}\prod_{j=1}^{k}(1+x_iy_j) = \sum_{\nu} s_\nu(x_1,\dots,x_m)\, s_{\nu'}(y_1,\dots,y_k),$$

the sum over the partitions ν with at most m parts and $\nu_1 \le k$. Let P be the probability measure $\frac{1}{k!}\prod_{i<j}|y_i-y_j|^2\,\frac{d\phi_1}{2\pi}\cdots\frac{d\phi_k}{2\pi}$ on the torus $y_j = e^{\mathrm{i}\phi_j}$. The Schur polynomials in k variables indexed by partitions with at most k parts are orthonormal for P: with $\delta = (k-1,\dots,0)$ and $a_\alpha(y) = \det[y_i^{\alpha_j}] = \sum_w \operatorname{sgn}(w)\, y^{w\alpha}$, one has $s_\mu = a_{\mu+\delta}/a_\delta$ and $|a_\delta|^2 = \prod_{i<j}|y_i-y_j|^2$, so $\int s_\mu \overline{s_\nu}\, dP = \frac{1}{k!}\int a_{\mu+\delta}\overline{a_{\nu+\delta}}\,\frac{d\phi}{(2\pi)^k}$; the monomials are orthonormal on the torus, and since the entries of $\mu+\delta$ and of $\nu+\delta$ are strictly decreasing, $y^{w(\mu+\delta)} = y^{w'(\nu+\delta)}$ only if $\mu=\nu$ and $w=w'$, which gives $\delta_{\mu\nu}$. Since $\nu \mapsto \nu'$ is injective,

$$s_\lambda(x) = \int \prod_{i,j}(1+x_iy_j)\,\overline{s_{\lambda'}(y)}\, dP(y),$$

and by the Cauchy–Schwarz inequality

$$|s_\lambda(x)| \le \sup_y \prod_{j=1}^{k}\prod_{i=1}^{m}|1+x_iy_j| \,\Big(\int |s_{\lambda'}|^2 dP\Big)^{1/2} = \Theta(x)^k. \qquad \square$$

Lemma 29 (Hardy estimate)**.** *Let $0 \le m \le n$, $k = n-m$, let $x_1,\dots,x_m \in (-1,1)$, let $r_1,\dots,r_m$ be distinct elements of $\{0,\dots,n-1\}$, and let h_* be holomorphic on a neighbourhood of the closed unit disk with $|h_*| \le K$ there. Then*

$$\Big|\det\big[x_i^{n-1-r_j} + h_*(x_i)x_i^{r_j}\big]_{i,j=1}^{m}\Big| \le V(x)\,(1+K)^m\,\Theta(x)^k\,2^k.$$

Proof. We follow the proof of [9, Lemma 4.1] and change two estimates. Put $Q(z) = \prod_i(1-x_iz)$, $P(z) = \prod_i(z-x_i)$, $B = P/Q$, and let $H_d = \{v/Q : \deg v < d\}$ with the norm of H^2 of the disk. As there, the orthogonal projection Π_m onto H_m preserves the values at the nodes, since the kernels $(1-x_iz)^{-1}$ lie in H_m; the reversal $R_n(v/Q) = z^{n-1}v(1/z)/Q$ is an isometry of H_n; and $L = R_n + h_*$ maps H_n to H^2 with norm at most $C_* = 1+K$. Let $u_j = z^j/Q$ for $0 \le j < n$, let $j_1 < \cdots < j_k$ be the indices not among the r's, and let $\ell_1(u),\dots,\ell_k(u)$ be the coordinates of $u \in H_n$ at $u_{j_1},\dots,u_{j_k}$. They are coefficients of the polynomial Qu, so by Parseval

$$\Big(\sum_{a=1}^{k}|\ell_a(u)|^2\Big)^{1/2} \le \|Qu\|_{H^2} \le \Theta\,\|u\|_{H^2}, \qquad \Theta = \max_{|z|=1}|Q(z)| = \Theta(x). \tag{1}$$

This is the first change: [9, (4.2)] bounds each $|\ell_a(u)|$ separately by $2^m\|u\|$. Define

$$Tu = T_1u + T_2u, \qquad T_1u = C_*^{-1}\Pi_m L u, \qquad T_2u = \Theta^{-1}\sum_{a=1}^{k}\ell_a(u)\,z^{a-1}B.$$

The functions $z^{a-1}B$, $1 \le a \le k$, are orthonormal because $|B| = 1$ on the circle, so (1) gives $\|T_2\| \le 1$; also $\|T_1\| \le 1$ and T_2 has rank at most k. This is the second change: by Weyl's inequality for singular values, $s_{i+j-1}(T_1+T_2) \le s_i(T_1) + s_j(T_2)$, applied with $j = k+1$, we have $s_{i+k}(T) \le s_i(T_1) \le 1$ for $i \ge 1$, and $s_i(T) \le \|T\| \le 2$ for $i \le k$. Hence

$$|\det T| = \prod_{i=1}^{n} s_i(T) \le 2^k,$$

in place of $|\det T| \le (k+1)^n$. The matrix of T from the input basis $(u_{r_1}, \dots, u_{r_m}$, then $u_{j_1}, \dots, u_{j_k})$ to the output basis $1/Q, \dots, z^{m-1}/Q, B, \dots, z^{k-1}B$ is block upper triangular: $T_2 u_{r_j} = 0$ and T_1 maps into H_m, and $T u_{j_a} = T_1 u_{j_a} + \Theta^{-1} z^{a-1} B$. Its diagonal blocks are $C_*^{-1}F$, with F the matrix of coordinates of $\Pi_m L u_{r_j}$ in $1/Q, \dots, z^{m-1}/Q$, and $\Theta^{-1} I_k$. The two bases have numerator coefficient matrices of determinant ± 1 with respect to the monomials, because P is monic of degree m. Hence $|\det F| = C_*^m \Theta^k |\det T| \le C_*^m \Theta^k 2^k$. Finally $L u_r = (z^{n-1-r} + h_* z^r)/Q$, its values at the nodes are preserved by Π_m, and $Q(x_i) > 0$; multiplying the values at x_i by $Q(x_i)$ shows that the matrix of the lemma is $[x_i^a]_{i \le m,\, a < m} F$, whose determinant is at most $V(x) C_*^m \Theta^k 2^k$. □

For nodes in $(-1, 1)$ the two lemmas give the bounds of [9, Lemma 4.1, Corollary 4.2] with $L_{n,m} = 2^{mk}(k+1)^n$ replaced by $\Theta(x)^k 2^k \le 2^{mk+k}$ and $\Theta(x)^k \le 2^{mk}$.

Lemma 30 (Far sheet)**.** *Let $0 \le m \le n$, $k = n - m$, let $r_1, \dots, r_m$ be distinct elements of $\{0, \dots, n-1\}$ and let $z_1, \dots, z_m$ be real. Then*

$$|\det[z_j^{r_i}]_{i,j=1}^m| \le m^{m/2}\, 2^{-\binom{m}{2}}\, 2^{mk/2} \prod_{j=1}^m (1+z_j^2)^{(n-1)/2}.$$

Proof. Write $z_j = \tan\phi_j$. Then $|z_j - z_i| = |\sin(\phi_j - \phi_i)|(1+z_i^2)^{1/2}(1+z_j^2)^{1/2}$, so $V(z) = \prod_j (1+z_j^2)^{(m-1)/2} \prod_{i<j} |\sin(\phi_j - \phi_i)|$, and $\prod_{i<j} |\sin(\phi_j - \phi_i)| = 2^{-\binom{m}{2}} \prod_{i<j} |e^{2\mathrm{i}\phi_j} - e^{2\mathrm{i}\phi_i}| \le 2^{-\binom{m}{2}} m^{m/2}$ by Hadamard's inequality. By Lemma 28 and $1 + |z| \le \sqrt{2}\,(1+z^2)^{1/2}$, $|\det[z_j^{r_i}]| \le V(z) \prod_j (1+|z_j|)^k \le V(z)\, 2^{mk/2} \prod_j (1+z_j^2)^{k/2}$. Since $(m-1) + k = n - 1$, the bound follows. □

Lemma 30 replaces $C_n 14^{nk}$, $C_n = n^{n/2} 2^{-\binom{n}{2}}$, in [9, Lemma 4.5]; after the weakening of $2^{-\binom{n}{2}}$ to $2^{-\binom{m}{2}}$ in the proof of [9, Lemma 6.2], and with $m^{m/2} \le n^{n/2}$, the factor 14^{nk} in [9, (6.6)] becomes $2^{mk/2}$.

Corollary 31 (Minors)**.** *Let the design be $(96, 15, 6, 8, 5)$ and put $\rho_1 = (2 - \alpha - \gamma_g)\log 2 - \Sigma_2 + 2\log 2$. Then $\rho_1 < 5.13821$, and for every $\varepsilon > 0$ there is N_ε such that for all $N \ge N_\varepsilon$, all $0 \le m \le n$ and all injective lists I, J of m row and column indices, with $k = n - m$,*

$$|\det A[I, J]| \le \exp\big((c_{\mathrm{up}} + \varepsilon) n^2 + \rho_1\, nk\big),$$

where $A[I, J]$ is the submatrix with rows I and columns J, and the empty minor equals 1.

Proof. We follow the proof of [9, Theorem 6.3], with Proposition 27 in place of [9, Proposition 5.4] and Lemmas 28–30 in place of [9, Lemmas 4.1, 4.5, Corollary 4.2]. The other ingredients hold for the design.

(1) Lemma 26 holds for every design: its proof uses only the integral representation of Section 2, which holds for every design [1, Sec. 5.1].

(2) Under the interpolation condition of [9, Lemma 6.2], $\sum_i (1 - x_i^2)/(1 + x_i^2) \le D_*$, as in Proposition 12, the row determinant is bounded through the entries $x^{n-1-r} + h_*(x) x^r$ with the function h_* of [9, Lemma 4.3]: if $n \ge 48$, $0 < D \le n$ and the nonzero nodes $x_i \in (-1, 1)$ satisfy $\sum_i (1 - x_i^2)/(1 + x_i^2) \le D$, there is a function h_*, holomorphic on a neighbourhood of the closed unit disk, with $h_*(x_i) = x_i^D$ for $x_i < 0$, $h_*(x_i) = -5 x_i^D$ for $x_i > 0$, and $|h_*| \le K_0 n$ there, $K_0 = 10 e^{12}$. [9, Lemma 4.3] needs $n \ge 48$, $0 < D_* \le n$ with $D_* = n - 1 - 2g$, and the sheet coefficients $c_\pm \in \{1, -5\}$, which come from the factor $\frac{3}{2}$ in the entries; here $n = 96N$ and $D_* = 80N - 1$, by (C1). Lemma 29 then gives [9, (6.5)] with $L_{n,m}$ replaced by $\Theta(x)^k 2^k \le 2^{mk+k}$. In the other case, the expansion over the sheets in the proof of [9, Lemma 6.2] bounds monomial determinants at real

nodes; with Lemma 30 in place of [9, Lemma 4.5], [9, (6.6)] holds with 14^{nk} replaced by $2^{mk/2}$. The column determinant is $\det[\psi_{c_b}(s_j)] = \prod_j s_j^b (1-s_j)^{q_0} \det[s_j^{c_b}]$, since $\psi_c(s) = s^{b+c}(1-s)^{q_0}$ [9, (2.11)], and Lemma 28 gives [9, (6.7)] with $L_{n,m}$ replaced by $\Theta(s)^k = \prod_j (1+s_j)^k \le 2^{mk}$. The constant $A_0 = 1 + K_0 + \frac{3}{2}$ does not involve the design.

(3) The factors of [9, (6.8)–(6.9)] become $(1+K_0 n)^m 2^k\, 2^{2mk}$ in the case of the interpolation condition and $(3/2)^m n^{n/2} 2^{3mk/2}$ in the other case, and $(1+K_0 n)^m$, $(3/2)^m n^{n/2}$ and 2^k are $e^{o(n^2)}$. Put $\theta = 1-t$ with $t = k/n$, so that $mk/n^2 = (1-t)t \le t$. By Proposition 27 and the definition of c_{up}, $\overline{U}_2(1) = U_2 + \frac{1}{2}\log 2 \le c_{\mathrm{up}}$, and

$$\overline{U}_2(1-t) - \overline{U}_2(1) = t\big(3\log 2 - (1+\alpha+\gamma_g)\log 2 - \Sigma_2\big) - \tfrac{3}{2}(\log 2)\, t^2,$$

so the first case gives the exponent at most $c_{\mathrm{up}} + \rho_1 t$. The second case gives at most

$$c_{\mathrm{up}} + (U_1 - U_2) + \big((2-\alpha-\gamma_g)\log 2 - \Sigma_1 - \lambda^{(1)}(1-2\gamma_g) + \tfrac{3}{2}\log 2\big)t.$$

The computation of Proposition 15 also bounds $\sup X$ and $\sup Y$ from below, since it encloses the values at the left ends of the brackets of the stationary points and the allowances $w \sup|X'|$ are below 3.7×10^{-22}; its table rounds upward at the fourteenth decimal. Hence $\Sigma_2 \ge -2.5750034368157$ and $\Sigma_1 \ge -2.9836915417496$; with $\alpha + \gamma_g = \frac{29}{96}$, $\gamma_g = \frac{1}{12}$, $\lambda^{(1)} = 2.71408765\ldots$ (Table 5) and $U_1 - U_2 < -0.1364$, the coefficients of t are below 5.13821 and 2.93858. Hence the first case dominates for all $t \in [0,1]$. □

5.6 The exponent

Lemma 32 (Perturbation; [9, Lemma 7.1])**.** *Let A, B be real $n \times n$ matrices and $R, S, K \ge 0$. If every $m \times m$ minor of A, including the empty one, has absolute value at most RS^{n-m}, and $|B_{ij}| \le K$, then $|\det(A+B)| \le R(1+nKS)^n$.*

Proof. Expand $\det(A+B)$ by choosing in each row the row of A or of B. A choice with k rows from B expands along those rows into at most n^k terms, each a product of k entries of B and a complementary minor of A of size $n-k$, so it is at most $n^k K^k R S^k$. Summing over the choices gives $R\sum_k \binom{n}{k}(nKS)^k = R(1+nKS)^n$. □

Definition 33 (Radius hypothesis)**.** For $\rho \ge 0$, hypothesis (R_ρ) is: as $p \to \infty$ through primes,

$$\sup_{|x-G| \le e^{-\rho n}} \log|\Delta_p(x)| \le (c_{\mathrm{up}} + o(1))\, n^2, \qquad n = n_0 p.$$

Proposition 34. *Assume (R_ρ) and put $\mu^* = \rho(1-\gamma_g+n_0\gamma)/\gamma$. For every $\varepsilon > 0$ there is b_0 such that $|G - a/b| > b^{-(\mu^*+\varepsilon)}$ for all integers a and $b \ge b_0$. In particular the irrationality exponent of G is at most μ^*.*

Proof. Put $\nu = 1-\gamma_g$, so that $n - g = \nu n$. Fix $\theta \in (0,1)$ and $\tau > 0$, and put $\sigma = \theta\gamma n_0/\nu + \tau$. Let b be large and $x = \nu \log b/(\theta\gamma n_0)$. By the prime number theorem the interval $[x, (1+\sigma)x]$ contains $(\sigma + o(1))x/\log x$ primes; at most $\log b/\log x = \theta\gamma n_0 x/(\nu \log x)$ of them divide b, and at most $|E_0|$ lie in E_0. Since $\sigma > \theta\gamma n_0/\nu$, for large b there is a prime p in this interval with $p > 4H_0$, $p \notin E_0$ and $p \nmid b$. Put $N = p$ and $n = n_0 p$, so that $\nu \log b/(\theta\gamma) \le n \le (1+\sigma)\nu \log b/(\theta\gamma)$. By Lemma 24, $\Delta_p(a/b) \ne 0$, and by Corollary 23 and $(n-g)\log b = \nu n \log b \le \theta\gamma n^2$,

$$\log|\Delta_p(a/b)| \ge (c_{\mathrm{low}} - \theta\gamma)n^2 - o(n^2) = \big(c_{\mathrm{up}} + (1-\theta)\gamma\big)n^2 - o(n^2).$$

If $|a/b - G| \le e^{-\rho n}$, then ($\mathrm{R}_\rho$) gives $\log|\Delta_p(a/b)| \le (c_{\mathrm{up}} + o(1))n^2$, which is impossible for large b. Hence $|G - a/b| > e^{-\rho n} \ge b^{-\rho\nu(1+\sigma)/(\theta\gamma)}$, and $\rho\nu(1+\sigma)/(\theta\gamma) \to \rho(\nu+n_0\gamma)/\gamma = \mu^*$ as $\theta \to 1$ and $\tau \to 0$. □

Theorem 35. *For the design* $(96, 15, 6, 8, 5)$ *put*

$$\rho_2 = \rho_1 + (\eta + \gamma_q)\log 2 + (1 - \gamma_g)\log(1 + \sqrt{2}), \qquad \mu_2 = \rho_2(1 - \gamma_g + n_0\gamma)/\gamma,$$

with ρ_1 *as in Corollary 31. For every* $\varepsilon > 0$ *there is* b_0 *such that* $|G - a/b| > b^{-(\mu_2+\varepsilon)}$ *for all integers* a *and* $b \geq b_0$*. Hence the irrationality exponent of* G *is at most* $\mu_2 < 2\,715$*.*

Proof. Let $\delta_1 > 0$ and $\rho = \rho_2 + \delta_1$. We prove (R_ρ); Proposition 34 then gives the bound with $\rho(1 - \gamma_g + n_0\gamma)/\gamma$ in place of μ_2, and δ_1 is arbitrary.

Let $\varepsilon > 0$, let p be a large prime, $n = n_0 p$, and $|x - G| \leq e^{-\rho n}$. Then $\Delta_p(x) = \det\big(A + (x - G)\Phi^{(1)}\big)$ with $A = \Phi^{(0)} + G\Phi^{(1)}$. By Corollary 31, every $m \times m$ minor of A is at most RS^{n-m} with $R = e^{(c_{\mathrm{up}}+\varepsilon)n^2}$ and $S = e^{\rho_1 n}$. By Lemma 25 and $d_r \leq n - 1 - g$, every entry of $(x - G)\Phi^{(1)}$ is at most $K = c_1 e^{-\rho n} e^{(\rho_2 - \rho_1)n + O(1)}$. Hence $nKS \leq c_1' n e^{-\delta_1 n}$, and Lemma 32 gives

$$|\Delta_p(x)| \leq e^{(c_{\mathrm{up}}+\varepsilon)n^2}\big(1 + c_1' n e^{-\delta_1 n}\big)^n = e^{(c_{\mathrm{up}}+\varepsilon+o(1))n^2}.$$

Since ε is arbitrary, (R_ρ) holds. For the design, $(\eta + \gamma_q)\log 2 + (1 - \gamma_g)\log(1 + \sqrt{2}) = \frac{11}{96}\log 2 + \frac{11}{12}\log(1 + \sqrt{2}) = 0.887349\ldots$, so $\rho_2 < 6.02556$; since $\rho_2((1 - \gamma_g)/\gamma + n_0)$ decreases in γ, $\gamma > 0.0025854$ and $1 - \gamma_g = \frac{11}{12}$, $\mu_2 < 6.02556\,(\frac{11}{12}/0.0025854 + 96) < 2\,715$. □

Remark 36. With the losses $L_{n,m} = 2^{mk}(k + 1)^n \leq 4^{nk}$ of [9, Lemma 4.1, Corollary 4.2] and 14^{nk} of [9, Lemma 4.5] in place of Lemmas 28–30, the coefficients of t in the proof of Corollary 31 are 6.52450 and 5.92421, the rate is 7.41185, and the same argument gives $\mu(G) < 3\,340$. With, in addition, the padding of [9, Proposition 5.4] in place of Proposition 27, the energy loses 2^{32n} per adjoined pair of nodes; for the design $(96, 15, 6, 8, 5)$ this holds because the design enters the padding only through $A + C + h \leq 2n$ and $2b + q \leq n$, which read $167N \leq 192N$ and $36N \leq 96N$. The total loss is then 2^{40nk}, the rate is $40\log 2 + 0.887 = 28.61$, and the same argument gives $\mu(G) < 12\,892$. For the design $(48, 7, 4, 4, 2)$, whose gap 1.303×10^{-4} is twenty times smaller, the padded rate 28.62 gives about 2.0×10^5. The bound 10^8 of [9] uses, in addition, the weakening of the loss to e^{1000nk}, the entry bound e^{3n} and the window $24000\log q < n \leq 72000\log q$. The bound 8 127 of [10], for the design $(48, 7, 4, 4, 2)$ with the gap 1.0×10^{-3} of its new certificate, uses a perturbation rate of 404.25 per unit of N, that is 8.42 per unit of n, against $\rho_2 = 6.03$ here; in it the energy lost per missing point is 5.37 per unit of n [10, Proposition 5.10], against the certified slope $\rho_1 - 2\log 2 = 3.75$ of Corollary 31.

Observation 37 (Measured; exact polynomials)**.** For the polynomials of Definition 19 we computed $\Delta_N(X)$ exactly (`measure_scout.py`, Appendix C), expanded it at G as $\sum_k c_k(X - G)^k$, and determined the largest d^* with $\sum_k |c_k|(d^*)^k \leq 2|c_0|$; then $|\Delta_N(x)| \leq 2|\Delta_N(G)|$ for $|x - G| \leq d^*$. The values of $\rho_n = -\log d^*/n$ for designs with the ratios $96 : 15 : 6 : 8 : 5$ and $48 : 7 : 4 : 4 : 2$, with the parameters rounded to integers where the ratio does not give integers (exact at $n = 96, 192$ for the first ratio and at $n = 48, 96$ for the second), are

n	48	72	96	120	144	192
$96 : 15 : 6 : 8 : 5$	4.442	4.474	4.520	4.541	4.551	4.551
$48 : 7 : 4 : 4 : 2$	4.454	4.539	4.566	4.534	–	–

(for the second ratio also $\rho_{24} = 4.326$).

The values ρ_n level off at 4.551; the rate proved in Theorem 35 is $\rho_2 = 6.03$. If (R_ρ) holds with $\rho = 4.6$, Proposition 34 gives an exponent at most 2.08×10^3 for the design $(96, 15, 6, 8, 5)$. For comparison, [8, Theorem 5.1] gives the exponent 24781 for $L(2, \chi_{-3})$, and [5, Corollary 1.2] the exponent 260 for $\zeta(5)$.

5.7 Structure near the constant

This subsection records measurements that account for the difference between the rate $\rho_2 = 6.03$ of Theorem 35 and the measured rate $\rho_n \approx 4.55$ of Observation 37. Throughout, $A = \Phi^{(0)} + G\Phi^{(1)}$ and $B = \Phi^{(1)}$.

Definition 38 (Singular values and pencil eigenvalues)**.** Let $\sigma_1 \geq \cdots \geq \sigma_n > 0$ be the singular values of A, and let u, v be unit vectors with $Av = \sigma_n u$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $A^{-1}B$, counted with multiplicity.

Lemma 39. *$\Delta_N(G + d) = \Delta_N(G) \prod_{i=1}^{n} (1 + d\lambda_i)$ for every real d. Hence the zeros of $\Delta_N(X)$ are the numbers $G - 1/\lambda_i$ with $\lambda_i \neq 0$, and the distance from G to the nearest zero is $1/\max_i |\lambda_i|$.*

Proof. $\Delta_N(G + d) = \det(A + dB) = \det A \, \det(I + dA^{-1}B)$, and $\det(I + dM) = \prod_i (1 + d\lambda_i)$ for the eigenvalues λ_i of Θ. □

Observation 40 (Measured; preprint ratio $48 : 7 : 4 : 4 : 2$)**.** The values of $\log \|A^{-1}\|_F / n$, computed in ball arithmetic, which bound $-\log \sigma_n / n$ from above within $\log n / n$, are

n	24	48	72	96	120	144
$\log \|A^{-1}\|_F / n$	4.103	4.257	4.341	4.373	4.353	4.352

(parameters rounded to integers for $n = 72, 120, 144$). At $n = 24$ and $n = 48$, $\log |u^{\mathsf{T}} B v| / n$ equals 0.221 and 0.197, and $-\log \sigma_n / n + \log |u^{\mathsf{T}} B v| / n$ equals 4.323 and 4.454; the measured ρ_n of Observation 37 are 4.326 and 4.454. The singular values of A decrease geometrically, with ratios between $e^{-3.3}$ and $e^{-6.6}$ at $n = 48$, and the zeros of $\Delta_N(X)$ lie at distances from G whose logarithms are in arithmetic progression with step 4.53 ($n = 24$) and 4.58 ($n = 48$).

These measurements indicate that the radius rate is the decay rate of the smallest singular value of A plus the size of B on the corresponding singular vectors, about 0.2; Theorem 35 uses the entry bound 0.887 for B.

Definition 41 (Energy at a given mass)**.** For $0 < \theta \leq 1$ let

$$F(\theta) = \max_{\mu, \nu} \Big(\theta \, L(\mu, \nu) + \theta^2 Q(\mu, \nu) \Big),$$

over probability measures μ on $[-1, 1]$ and ν on $[0, 1]$, where $L(\mu, \nu) = (1 + \alpha + \gamma_g) \log 2 + \int W_2 \, d\mu + \int W_s \, d\nu$ and

$$\begin{aligned} Q(\mu, \nu) = \tfrac{1}{2} \log 2 &+ \iint \log |x - x'| \, d\mu \, d\mu + \tfrac{1}{2} \iint \log(1 - xx') \, d\mu \, d\mu \\ &+ \iint \log |s - s'| \, d\nu \, d\nu - \iint \log(1 - 2xs + x^2) \, d\mu \, d\nu, \end{aligned}$$

with W_2, W_s as in Section 6. The terms of L come from one node each and those of Q from pairs of nodes, as in the proof of Proposition 27, and $F(1) - \tfrac{1}{2} \log 2 = U_2^*$ (Proposition 45). For a maximising pair (μ^*, ν^*) at $\theta = 1$ put

$$r_{\mathrm{row}} = \max_{|z| = 1} \int \log |1 + xz| \, d\mu^*(x), \qquad r_{\mathrm{col}} = \max_{|z| = 1} \int \log |1 + sz| \, d\nu^*(s).$$

By the expansions in the proof of Proposition 27, $Q(\mu,\nu)$ is $-\frac{3}{2}\log 2$ minus a sum of squares of moments, as in Definition 44, so Q is a concave quadratic form in the pair of measures. With $\tilde\mu=\theta\mu$ and $\tilde\nu=\theta\nu$, $F(\theta)$ is the maximum of a concave function of $(\theta,\tilde\mu,\tilde\nu)$ over a convex set, so F is concave and $F(1-t)\le F(1)-tF'(1)$ for $0\le t<1$. The bound $\overline{U}_2(\theta)$ of Proposition 27 has the form $\theta L+\theta^2 Q$, with the certificate of (C4) in place of a pair of measures. Heuristically, the largest energies of configurations of $m=\theta n$ nodes are $F(\theta)n^2(1+o(1))$, so $-F'(1)$ is the smallest slope at $\theta=1$ that a bound of this kind can have. Removing one row and one column of a monomial alternant on nodes $x_1,\dots,x_m$ multiplies it by an elementary symmetric function of the nodes, whose maximum over the removed exponent is of order $e^{m\max_{|z|=1}\int\log|1+xz|\,d\mu}$; this is the origin of r_{row} and r_{col}.

Observation 42 (Numerical; discretised equilibrium problem)**.** With the solver of Section 6 on a grid of 300×200 cells, $F(1)$ reproduces the values of that section, and

design	$-F'(1)$	r_{row}	r_{col}	sum
$(48,7,4,4,2)$	3.766	0.178	0.434	4.378
$(96,15,6,8,5)$	3.752	0.177	0.438	4.367
$(48,7,4,8,2)$	3.844	0.187	0.435	4.467
$(48,14,4,4,2)$	3.953	0.188	0.457	4.598

The derivative was computed by the envelope formula $F'(1)=L(\mu^*,\nu^*)+2Q(\mu^*,\nu^*)$ and agrees with finite differences (3.748 at $\theta=0.99$ for the first design). For the design $(96,15,6,8,5)$, $-F'(1)=3.752$ agrees with the certified coefficient $\rho_1-2\log 2=3.7519$ of Corollary 31.

Conjecture 43. For a design satisfying (C1)–(C4), $\lim_{N\to\infty}-\log\sigma_n/n=-F'(1)+r_{\mathrm{row}}+r_{\mathrm{col}}$.

For the design of the preprint the right side is 4.378, and Observation 40 gives 4.35 at $n=96,120,144$. The rates therefore split into three parts. In Theorem 35, $\rho_2=6.03$ is the energy slope 3.75, the bound $2\log 2=1.39$ of Lemmas 28 and 29 for the row and column determinants, and the entry bound 0.89 for B. In the measurements, $\rho_n\approx4.55$ is the same slope 3.75, the alternant terms $r_{\mathrm{row}}+r_{\mathrm{col}}=0.61$, and the size 0.2 of B on the bottom singular vectors. With the measured parts in place of the proved ones, the argument of Theorem 35 would give an exponent near $(4.37+0.2)(1-\gamma_g+n_0\gamma)/\gamma\approx2.1\times10^3$ for the design $(96,15,6,8,5)$. Lemmas 28 and 29 give the alternant terms for each configuration, as $\frac{1}{n}\log\Theta(x)$ and $\frac{1}{n}\log\Theta(s)$; the bound $2\log2$ comes from bounding these by $\log2$ before the energy is maximised. Keeping them inside the energy, with a certificate for the resulting problem, would give the slope $-F'(1)+r_{\mathrm{row}}+r_{\mathrm{col}}=4.367$ in place of 5.138 and an exponent near 2.5×10^3; the rest of the difference would need a bound for B on the bottom singular vectors of A.

6 The best constant of the real-place method

The constant U_κ of Definition 5 depends on the trial sequences. Here we identify its infimum. Let

$$W_\kappa(x)=(\alpha+2\gamma_g)\log|x|+2\eta\log(1-x)-\Big(\frac{\kappa}{2}+\alpha+\eta+\gamma_g\Big)\log(1+x^2),\quad W_s(s)=\beta\log s+\gamma_q\log(1-s),$$

and $D(x)=2\gamma_g-2x^2/(1+x^2)$, with the values $-\infty$ at $x=0$, $x=1$, $s=0$, $s=1$; these functions are upper semicontinuous and bounded above. Then $X_{p,v,\lambda}=W_\kappa+\lambda D-2\kappa T(p,\cdot)-S(v,\cdot)$, and in Definition 5 the suprema may be taken over the closed intervals $[-1,1]$ and $[0,1]$ without changing their values. Let

$$U_\kappa^*=\inf\{U_\kappa(p,v,\lambda)\},$$

the infimum over admissible p, v and $\lambda \in \Lambda_\kappa$. By Proposition 12, $\max(U_1^*, U_2^*)$ is the best upper constant the method of [1, Sec. 6] can give for the design.

Definition 44 (Moments and energy)**.** For a Borel probability measure μ on $[-1,1]$ and ν on $[0,1]$ put

$$A_k = \int T_k \, d\mu, \qquad B_k = \int x^k \, d\mu(x), \qquad C_k = \int T_k \, d\nu \qquad (k \geq 1),$$

and

$$\Phi_\kappa(\mu,\nu) = (-1+\alpha+\gamma_g)\log 2 + \int W_\kappa \, d\mu + \int W_s \, d\nu - \kappa \sum_{k\geq 1} \frac{A_k^2}{k} - \frac{1}{2}\sum_{k\geq 1} \frac{(B_k - 2C_k)^2}{k}.$$

The integrals lie in $[-\infty, \infty)$ and the two sums in $[0,\infty]$, so $\Phi_\kappa(\mu,\nu) \in [-\infty,\infty)$ is well defined. Let $\mathcal{M}_2$ be the set of all pairs (μ,ν), and $\mathcal{M}_1$ the set of pairs with $\int D \, d\mu \geq 0$.

Proposition 45. *For $\kappa \in \{1,2\}$,*

$$U_\kappa^* = \max_{(\mu,\nu)\in\mathcal{M}_\kappa} \Phi_\kappa(\mu,\nu),$$

and the maximum is attained. Moreover, let (μ^*,ν^*) be any maximising pair, with moments A_k^*, B_k^*, C_k^*. Then for all admissible p, v and $\lambda \in \Lambda_\kappa$,

$$U_\kappa(p,v,\lambda) - U_\kappa^* \;\geq\; \kappa \sum_{k\geq 1} \frac{(p_k - A_k^*)^2}{k} + \frac{1}{2}\sum_{k\geq 1} \frac{(v_k - B_k^* + 2C_k^*)^2}{k}. \tag{2}$$

In particular all maximising pairs have the same sequences (A_k^*) and $(B_k^* - 2C_k^*)$.

Proof. Step 1 (a lower bound for every certificate). Fix admissible p, v, $\lambda \in \Lambda_\kappa$ and $(\mu,\nu) \in \mathcal{M}_\kappa$. Since a supremum dominates an average,

$$\sup X_{p,v,\lambda} \geq \int X_{p,v,\lambda} \, d\mu, \qquad \sup Y_v \geq \int Y_v \, d\nu.$$

By uniform convergence, $\int T(p,x)\,d\mu = \sum_k p_k A_k/k$, $\int S(v,x)\,d\mu = \sum_k v_k B_k/k$ and $\int T(v,s)\,d\nu = \sum_k v_k C_k/k$. When $\kappa = 1$, $\lambda \int D \, d\mu \geq 0$. Hence

$$U_\kappa(p,v,\lambda) \geq (-1+\alpha+\gamma_g)\log 2 + \int W_\kappa \, d\mu + \int W_s \, d\nu + \sum_{k\geq 1} \frac{\kappa p_k^2 - 2\kappa p_k A_k}{k} + \sum_{k\geq 1} \frac{\frac{1}{2}v_k^2 - v_k(B_k - 2C_k)}{k}.$$

The sums converge absolutely because p, v are admissible and $|A_k|, |B_k|, |C_k| \leq 1$. Completing the squares,

$$U_\kappa(p,v,\lambda) \geq \Phi_\kappa(\mu,\nu) + \kappa \sum_k \frac{(p_k - A_k)^2}{k} + \frac{1}{2}\sum_k \frac{(v_k - B_k + 2C_k)^2}{k}, \tag{3}$$

where, if $\Phi_\kappa(\mu,\nu) = -\infty$, the inequality holds trivially, and otherwise the two sums $\sum A_k^2/k$, $\sum (B_k - 2C_k)^2/k$ are finite and the rearrangement is legitimate. In particular $U_\kappa^* \geq \sup_{\mathcal{M}_\kappa} \Phi_\kappa$.

Step 2 (truncated sequences and a minimax theorem). Fix $K \geq 1$ and $M > 0$. Let $W_\kappa^M = \max(W_\kappa, -M)$ and $W_s^M = \max(W_s, -M)$; these are continuous real functions. Write $U_\kappa^{K,M}(p,v,\lambda)$ for U_κ with W_κ, W_s replaced by W_κ^M, W_s^M, restricted to sequences with $p_k = v_k = 0$ for $k > K$, which we identify with $(p,v) \in \mathbb{R}^{2K}$. For (μ,ν) let

$$L(p,v,\lambda;\mu,\nu) = (-1+\alpha+\gamma_g)\log 2 + \kappa\|p\|_*^2 + \tfrac{1}{2}\|v\|_*^2 + \int X_{p,v,\lambda}^M \, d\mu + \int Y_v^M \, d\nu,$$

with X^M, Y^M built from the truncated weights. Since X^M and Y^M are continuous on compact intervals, $\sup_{\mu,\nu} L = U_\kappa^{K,M}(p, v, \lambda)$, the supremum being attained at point masses. The function L is real valued, convex and continuous in $(p, v, \lambda) \in \mathbb{R}^{2K} \times \Lambda_\kappa$, and affine and weakly continuous in (μ, ν), which ranges over a convex set that is compact in the weak topology. Sion's minimax theorem [4] gives

$$\inf_{p,v,\lambda} U_\kappa^{K,M} = \sup_{\mu,\nu} \inf_{p,v,\lambda} L.$$

The inner infimum over (p, v) is a finite-dimensional quadratic minimisation with minimiser $p_k = A_k$, $v_k = B_k - 2C_k$ $(k \le K)$; the infimum over $\lambda \in \Lambda_1$ of $\lambda \int D\, d\mu$ is 0 if $\int D\, d\mu \ge 0$ and $-\infty$ otherwise. Therefore

$$\inf_{p,v,\lambda} U_\kappa^{K,M} = \sup_{\mathcal{M}_\kappa} \Phi_\kappa^{K,M},$$

where $\Phi_\kappa^{K,M}$ is Φ_κ with W_κ^M, W_s^M in place of W_κ, W_s and both sums truncated at $k \le K$.

Step 3 (limits). Since $W^M \ge W$, $U_\kappa^{K,M} \ge U_\kappa$ on truncated sequences, which are admissible. Hence

$$U_\kappa^* \le \inf_{p,v,\lambda} U_\kappa^{K,M} = \sup_{\mathcal{M}_\kappa} \Phi_\kappa^{K,M} \qquad \text{for all } K, M.$$

Take $M = K$ and write $\Phi^{(K)} = \Phi_\kappa^{K,K}$. Each $\Phi^{(K)}$ is weakly continuous on the compact set $\mathcal{M}_\kappa$ (closed because D is continuous), so it attains its supremum at some (μ_K, ν_K). Pass to a subsequence converging weakly to $(\mu^*, \nu^*) \in \mathcal{M}_\kappa$. The functionals decrease in K: for $J \le K$, $\Phi^{(K)} \le \Phi^{(J)}$, because the truncated weights decrease and the subtracted sums increase. Thus, for fixed J,

$$\limsup_{K\to\infty} \Phi^{(K)}(\mu_K, \nu_K) \le \limsup_{K\to\infty} \Phi^{(J)}(\mu_K, \nu_K) = \Phi^{(J)}(\mu^*, \nu^*).$$

As $J \to \infty$, $\int W_\kappa^J\, d\mu^* \downarrow \int W_\kappa\, d\mu^*$ and $\int W_s^J\, d\nu^* \downarrow \int W_s\, d\nu^*$ by monotone convergence (the weights are bounded above), and the truncated sums increase to the full sums. Hence $\Phi^{(J)}(\mu^*, \nu^*) \downarrow \Phi_\kappa(\mu^*, \nu^*)$, and

$$U_\kappa^* \le \Phi_\kappa(\mu^*, \nu^*) \le \sup_{\mathcal{M}_\kappa} \Phi_\kappa \le U_\kappa^*,$$

the last inequality by Step 1. All are equalities and the maximum is attained at (μ^*, ν^*).

Step 4 (excess inequality and uniqueness of the moments). Apply (3) with $(\mu, \nu) = (\mu^*, \nu^*)$ and use $\Phi_\kappa(\mu^*, \nu^*) = U_\kappa^*$; this is (2). Let $(p^{(j)}, v^{(j)}, \lambda^{(j)})$ be admissible with $U_\kappa \to U_\kappa^*$. By (2) applied to two maximising pairs, $p^{(j)}$ converges in the weighted norm $\|\cdot\|_*$ to the A-sequence of each of them, so these sequences coincide; likewise for $B - 2C$. □

Remark 46 (Logarithmic form)**.** If $\iint \log|x - x'|\, d\mu\, d\mu$, $\iint \log|s - s'|\, d\nu\, d\nu$ and $\iint \log(1 - 2xs + x^2)\, d\mu\, d\nu$ are finite, then

$$\begin{aligned}\Phi_\kappa(\mu, \nu) = \Big(\alpha + \gamma_g + \frac{\kappa}{2}\Big) \log 2 + \int W_\kappa\, d\mu + \int W_s\, d\nu + \frac{\kappa}{2} \iint \log|x - x'|\, d\mu\, d\mu \\ + \frac{1}{2} \iint \log(1 - xx')\, d\mu\, d\mu + \iint \log|s - s'|\, d\nu\, d\nu - \iint \log(1 - 2xs + x^2)\, d\mu\, d\nu.\end{aligned}$$

This follows from the expansions [1, Eqs. (81), (82)]: integrate them with a damping factor $r < 1$, where the series converge uniformly, and let $r \uparrow 1$. The damped sums $\sum r^k A_k^2/k$ increase to $\sum A_k^2/k$, and the damped kernels converge by dominated convergence, using $|1 - r\zeta|^2 \ge r|1 - \zeta|^2$ for $|\zeta| = 1$ as in [1, Sec. 6]. Formally, Φ_κ evaluated at the empirical measures of a configuration is the $n \to \infty$ form of the exact identity [1, Eq. (80)]. We do not prove here that $\sup n^{-2} \log I_{\kappa,n}$ converges to $\max \Phi_\kappa$; if it does, the method of the preprint loses nothing after the passage from $|\Delta_N|$ to the integrands $I_{\kappa,n}$.

Remark 47 (The constraint for $\kappa = 2$)**.** On the configurations treated with $\kappa = 2$, condition [1, Eq. (67)] gives $\langle D \rangle \le -1/n$, so $\lambda \le 0$ is also admissible there; in the dual this adds the constraint $\int D\, d\mu \le 0$. At the maximisers computed below $\int D\, d\mu^* \approx -0.34$, so the constraint is inactive and the preprint loses nothing by taking $\lambda = 0$.

6.1 Numerical solution

The script `cat_equilibrium.py` maximises Φ_κ over pairs of measures with piecewise-constant densities on Chebyshev-spaced cells (500 cells for μ, 330 for ν), using the logarithmic form of Remark 46 with exact cell integrals of $\log|x-y|$ and of the logarithmic weights. By Step 1, every value obtained this way is a lower bound for U_κ^*; every certificate gives an upper bound.

Observation 48 (Numerical)**.** The run of `cat_equilibrium.py` of 7 October 2026 gives the following.

1. For the preprint's design $(48, 7, 4, 4, 2)$: $-2.292188 \le U_2^* \le -2.291920$, where the upper value comes from a certificate with 300 free coefficients. The dual value for $\kappa = 1$ is $-2.43701 \le U_1^*$, with no matching certificate computed; the preprint's own $\kappa = 1$ certificate gives -2.29679. The finite-place constant of [1, Prop. 3.4] is -2.290810. The largest margin the method can certify for this design lies in $[0.00111, 0.00138]$; the preprint certifies 0.00013.

2. For the design $(96, 15, 6, 8, 5)$: $-2.278235 \le U_2^* \le -2.278134$ and $-2.41499 \le U_1^* \le -2.41461$, the upper values coming from the certificate of Proposition 15; finite-place constant -2.275548, largest certifiable margin in $[0.00259, 0.00269]$.

3. The supports of the maximising measures are close to $[-1, -0.100] \cup [0.100, 0.993\text{–}0.995]$ for μ and $(0.006, 1)$ for ν, for both designs.

4. The moments A_k^* and $B_k^* - 2C_k^*$ for the preprint's design agree with the trial sequences printed in [1, Sec. 7] to within 3.2×10^{-3} for $k \le 10$.

Item 4 is what (2) predicts: the preprint's $\kappa = 2$ certificate exceeds U_2^* by 0.0010–0.0013, so its sequences lie within that weighted squared distance of the moments of the equilibrium measures. The trial coefficients of the preprint are approximations of these moments, and the geometric tails it uses, with bases near 1 and near $\pm 0.1 + 0.95i$, reflect the endpoints of the supports, in particular the gap around $x = 0$. The certificate of [10, Proposition 5.11], checked in Lean, gives $U_2 < -2.29180925$ for the design $(48, 7, 4, 4, 2)$, that is a gap of 1.0×10^{-3}. The numerical certificate with 300 free coefficients of item 1 gives $U_2 \le -2.291920$, lower by 1.1×10^{-4}, so larger trial sequences still change U_2 in the fourth decimal (compare [10, Remark 6.1]).

Observation 49 (Numerical; design $(96, 15, 6, 8, 5)$, $\kappa = 2$)**.** On a mesh refined toward the support edges (688 cells for μ, 279 for ν) the dual value is -2.2782349, within 10^{-6} of the value on the regular grid. The supports are $[-1, -0.1011] \cup [0.0981, 0.9932]$ for μ and $[0.0065, 0.99992]$ for ν. Near $x = -1$ the density of μ behaves like $(1+x)^\theta$ with fitted $\theta = -0.340$; at the two edges of the gap it vanishes like a power with fitted exponents 0.475 and 0.462; at the lower edge of ν it rises like a square root.

Remark 50 (The edge at $x = -1$; heuristic)**.** Near $x = -1$, with $u = 1 + x$, the kernel $\frac{\kappa}{2}\log|x - x'| + \frac{1}{2}\log(1 - xx')$ is $\frac{\kappa}{2}\big(\log|u - u'| + \frac{1}{\kappa}\log(u + u')\big) + O(uu')$, and the external field is smooth. For a local density u^θ on $(0, \infty)$ with $-1 < \theta < 0$ one has $\mathrm{PV}\int_0^\infty \frac{t^\theta}{u - t}\, dt = \pi u^\theta \cot(\pi\theta)$ and $\int_0^\infty \frac{t^\theta}{u+t}\, dt = -\frac{\pi u^\theta}{\sin(\pi\theta)}$, so the leading singular part of the derivative of the potential vanishes exactly when $\cos(\pi\theta) = 1/\kappa$. For $\kappa = 2$ this gives $\theta = -1/3$, in agreement with Observation 49; for $\kappa = 1$ it gives a bounded density. We have no proof of this local analysis.

Code and data

The scripts, their logs, the coefficient files and the source of this note are at `https://github.com/MarcosCarreira/catalan-designs`. The repository also contains the verification report [3], the scripts that check the design of [1], and the scripts of the companion note [12] on $\beta(4)$.

Acknowledgement

This note was prepared with the assistance of Claude (Anthropic). All computations were run by the author on his own computer; the logs are listed in the appendix.

References

[1] OpenAI, *Catalan's constant is irrational*, preprint, 24 September 2026, `https://github.com/openai/math/blob/main/preprints/Catalans-constant-is-irrational-September-24-2026/paper.pdf`.

[2] OpenAI, *Irrationality of Catalan's constant*, Lean 4 formalisation, repository `openai/math`, scope note `https://github.com/openai/math/blob/main/lean/docs/005.md`; formal statement in `lean/ComparatorChallenges/Catalan.lean`.

[3] M. Costa Santos Carreira, *Verification of the preprint "Catalan's constant is irrational" (OpenAI, 24 September 2026)*, report, 7 October 2026, sent to the organizers of the Ramanujan Challenge; included in the repository `https://github.com/MarcosCarreira/catalan-designs`.

[4] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.

[5] A. Fauzan, $\zeta(5)$ *is irrational*, preprint, Zenodo 22826419, 17 September 2026.

[6] Z.-W. Sun, *Catalan's constant is irrational*, preprint, arXiv:2609.04176, 3 September 2026.

[7] D. Wachs, *A note on a recent claimed proof of the irrationality of Catalan's constant*, preprint, arXiv:2609.22339, 16 September 2026.

[8] F. Calegari, V. Dimitrov and Y. Tang, *Arithmetic holonomy bounds and effective Diophantine approximation*, preprint, arXiv:2510.04156, 2025.

[9] R. K. Jha, *A finite irrationality measure for Catalan's constant*, preprint, Zenodo, 8 October 2026, doi:10.5281/zenodo.23237269; Lean sources at `https://github.com/rohitkrjha/catalan-measure`.

[10] S. Litsey, *An irrationality measure for Catalan's constant*, preprint, 10 October 2026, with a Lean formalization, `https://github.com/slitsey/catalan-irrationality-measure`.

[11] I. G. Macdonald, *Symmetric functions and Hall polynomials*, 2nd ed., Oxford University Press, 1995.

[12] M. Costa Santos Carreira, *Determinant constructions for the Dirichlet beta value at four*, preprint, 9 October 2026, `https://github.com/MarcosCarreira/catalan-designs`.

Appendices

Appendix A. Pivots modulo 131

Table 3 lists the pivots of the Gaussian elimination of Proposition 14, in the layout of [1, Table 1]: rows $R_i = B(i,\cdot) + \sigma B(i+1,\cdot)$ in increasing order, a swap with the first later row having a nonzero entry in column i when the diagonal entry vanishes, and elimination below the pivot. All 3×96 pivots are nonzero.

σ	Successive pivots in $\mathbb{F}_{131}$
0	89, 55, 124, 94, 75, 121, 36, 40, 20, 2, 34, 28, 11, 118, 31, 8
	73, 28, 32, 1, 121, 86, 91, 100, 14, 119, 20, 51, 78, 23, 5, 118
	40, 128, 109, 56, 19, 9, 93, 59, 90, 2, 115, 104, 36, 89, 106, 85
	89, 106, 20, 121, 89, 20, 11, 126, 42, 18, 127, 43, 59, 35, 6, 19
	80, 111, 126, 23, 1, 6, 110, 29, 101, 94, 38, 104, 62, 100, 10, 77
	111, 60, 91, 83, 61, 39, 6, 124, 20, 73, 14, 25, 62, 57, 81, 42
1	43, 52, 97, 6, 12, 101, 74, 50, 67, 120, 18, 28, 41, 68, 17, 92
	53, 130, 104, 9, 120, 27, 112, 43, 63, 89, 17, 8, 114, 73, 22, 127
	20, 81, 87, 6, 31, 32, 29, 112, 129, 78, 59, 70, 119, 125, 128, 108
	95, 47, 31, 72, 43, 71, 32, 105, 130, 2, 104, 129, 65, 114, 104, 39
	99, 122, 41, 56, 124, 71, 17, 29, 59, 61, 41, 112, 9, 68, 12, 22
	56, 109, 67, 103, 70, 123, 1, 70, 9, 35, 79, 67, 29, 84, 62, 39
-1	4, 70, 20, 114, 56, 52, 14, 19, 50, 7, 22, 8, 82, 125, 8, 21
	8, 14, 124, 103, 28, 59, 78, 114, 39, 66, 14, 62, 45, 28, 74, 123
	24, 78, 19, 38, 112, 101, 68, 41, 2, 41, 111, 11, 124, 35, 33, 13
	58, 9, 15, 122, 109, 113, 103, 82, 93, 73, 99, 86, 102, 117, 40, 111
	102, 44, 99, 21, 95, 20, 76, 102, 42, 95, 62, 4, 78, 16, 108, 130
	24, 62, 11, 87, 108, 9, 22, 20, 44, 74, 83, 83, 53, 26, 27, 42

Table 3: Gaussian pivots of $B_0 + \sigma B_1$ modulo 131 for the design $(96, 15, 6, 8, 5)$, computed by the elimination rule of Section 4.4 of the preprint (rows in increasing order, indices from zero). Row swaps: $\sigma = 0$: $(71, 72)$; $\sigma = 1$: none; $\sigma = -1$: $(3, 4)$.

Appendix B. Trial coefficients

Tables 4 and 5 give the coefficients of Proposition 15: $l_1, \ldots, l_{16}$, then r_z for the real bases, then a_j, b_j for the complex bases z_j of Table 2, as in Section 4.4. They are the contents of the file `cat_tail_edges_out.json` read by `cat_cert7_new.py`, whose SHA-256 is

`f9466e8b08e882ef93f7d6738e091e190890a4644109f4f7bff3cb2f047f08fb`.

Table 4: Exact trial coefficients for $\kappa = 2$ (design $(96, 15, 6, 8, 5)$, bases of Section 4.4; a_j, b_j belong to the complex base z_j of Table 2). Every entry is a terminating decimal, equal to the rational number with denominator dividing 10^{20} used by the certificate.

coefficient	p	v
l_1	0.206800750042179584	−0.79574677215463866368
l_2	−0.28212239555494916096	−1.4989422266403733504
l_3	−0.02113163454145066624	2.5009964709022875648
l_4	0.05789817505804435456	−0.81823277725023289344
l_5	0.00026235849900718592	−0.34335767523865288704
l_6	−0.0072217607220025536	−0.320596670735328256
l_7	−0.00077986273687198576	0.60074342766141300736
l_8	0.00583190639606234112	−0.22903838638739832832
l_9	−0.00019523630604845356	−0.09995757566704955392
l_{10}	−0.00007169915474324479	−0.04811003309347750912
l_{11}	−0.00000108655493200221	0.12107402459375419392
l_{12}	0.00105108057059437104	−0.05769231465011724288
l_{13}	−0.00000693175149893269	−0.01617537493579744512
l_{14}	0.00001725288066305452	−0.00372167680747767168
l_{15}	0.00000973599251857202	0.01588745016032467456
l_{16}	0.00014102882971544792	−0.01034612270527625216
$r_{-0.7}$	−0.04786916011755584512	3.37330581687454662656
$r_{-0.85}$	−0.00338598655255290304	−0.12404277748026001408
$r_{-0.93}$	−0.00152976622486150048	0.04885985700061149184
$r_{-0.97}$	−0.00120675674217382288	0.00138138687187643376
$r_{-0.99}$	0.00019165391468863168	0.01164792422081226624
$r_{-0.997}$	−0.00007907329481067227	−0.00229561757343049248
$r_{-0.999}$	0.00002553545189618694	0.00473646688713458944
$r_{0.5}$	0.03373230427090172928	−0.50180710810041892864
$r_{0.8}$	0.00013224682015582930	0.38923514859669872640
$r_{0.9}$	−0.11220349712156209152	−0.21185139014654906368
$r_{0.95}$	0.04111662498598798848	0.11820319904626905088
$r_{0.98}$	−0.01903291227711012096	−0.12476878794932975616
$r_{0.99}$	0.03093667908251600896	−0.02728980612827940864
$r_{0.993}$	−0.03510075192808729600	0.56321313544565563392
$r_{0.995}$	0.01548609449841733632	−0.46525289648086761472
$r_{0.998}$	−0.00198049920931303680	0.10953924526435145728
$r_{0.999}$	0.00071932905135042120	−0.04621039255742801920
$r_{0.9995}$	−0.00013994365914259064	0.00962170839594888320
a_1	0.01378419297512573184	−0.21918082265604956160
b_1	0.02126164990680664064	−1.33809826164443365376
a_2	−0.01733763451901569024	−0.01711531054182729984
b_2	0.03029540697900915200	−0.08296597224054435840
a_3	−0.00756011835073578368	−0.00391227782152036096

coefficient	p	v
b_3	0.0130093944140507084 8	0.0018905765643601952
a_4	−0.0021980954774932339 2	0.0003655361447248898 4
b_4	0.0024908364160451273 6	−0.0000319392882983672 7
a_5	−0.0004487362422144022 4	−0.0000503789425718966 9
b_5	0.0005140007036752244 8	0.0000042284852621798 1
a_6	0.0093645283370664793 6	2.9289960087510827008
b_6	−0.022923074027861504	−0.73530769082054770688
a_7	−0.0150214890673654169 6	0.22470588141675986944
b_7	−0.0282205335411368908 8	0.29274714266140844032
a_8	−0.006989155217107808	−0.03656654427860699648
b_8	−0.0118833407452262579 2	−0.0035922647932014534 4
a_9	−0.0019533767804755203 2	0.0041998114478016121 6
b_9	−0.0023332790194249088	−0.00074394748896074352
a_{10}	−0.0004199987532926987 2	−0.00056237857865506688
b_{10}	−0.0004429997096296190 4	0.00013910295345217162
a_{11}	−0.0095372679916514764 8	0.0058534892524637568
b_{11}	−0.0038796137470807788 8	0.0102808701917307827 2
a_{12}	−0.0045122442652621171 2	−0.0036099657002659852 8
b_{12}	−0.0035332379215008748 8	−0.0011832660350377676 8
a_{13}	0.0004709432710726769 6	0.0013352270534496755 2
b_{13}	−0.001142595745546441 6	0.0001899920182740598 4
a_{14}	−0.0004167619462085335 2	−0.0001522352932889825 6
b_{14}	−0.0000603936805912453 2	−0.0000053138955337855 8
a_{15}	−0.1922074450940608102 4	−4.0108480434345495756 8
b_{15}	−0.0027233399860160617 6	5.3173862987596215091 2
a_{16}	0.001987967941366850 4	0.5992801124451046195 2
b_{16}	0.0041014904003939692 8	−0.4005880067966764646 4
a_{17}	0.0005112700327553320 8	−0.1211182566658550784
b_{17}	−0.0007936226501798041 6	−0.102845665185202176
a_{18}	−0.0000638810130163678 4	−0.0054667629090955340 8
b_{18}	0.0000951155752229426 4	−0.0490092935013012582 4
a_{19}	0.0000082263038945635 2	0.0080829708335681523 2
b_{19}	−0.0000126888617025378 2	−0.0232056942074883328

Table 5: Exact trial coefficients for $\kappa = 1$ (design $(96, 15, 6, 8, 5)$, bases of Section 4.4; a_j, b_j belong to the complex base z_j of Table 2). Every entry is a terminating decimal, equal to the rational number with denominator dividing 10^{20} used by the certificate. The multiplier is $\lambda = 2.71408765245248012288$.

coefficient	p	v
l_1	−1.3643182188079734784	−0.6603368434820706304
l_2	−12.043081600007465861 12	2.0680447339678266163 2
l_3	37.0067935806623842304	−1.2863113017592828723 2
l_4	−26.333925024799580487 68	−0.0373754438396064665 6
l_5	−3.6881418263488862617 6	0.0834900120242207232
l_6	−0.6678009620559967027 2	0.454161895431460864
l_7	9.2811692064010469376	−0.3566412053181928243 2
l_8	−7.5662610947167066521 6	0.0101145793704035673 6
l_9	−1.5543871076032023756 8	0.0461614347376700416
l_{10}	0.4314040825898136371 2	0.0799507030844381184

coefficient	p	v
l_{11}	1.98619553818667122688	−0.08064489931389446144
l_{12}	−1.94221592705101824	0.01284833796006078464
l_{13}	−0.28539702152857997312	0.00955929775567621504
l_{14}	0.18794623949996654592	0.00830588207825044992
l_{15}	0.22014073045237035008	−0.01193608321753916416
l_{16}	−0.38797524471677739008	0.00459182415073425728
$r_{-0.7}$	58.36565850394278756352	−2.06299007198147739648
$r_{-0.85}$	−2.01707726563734454272	0.08828065155060977664
$r_{-0.93}$	0.245413203317723136	−0.01207819266465484032
$r_{-0.97}$	−0.0483495181067449344	0.00252936410576541376
$r_{-0.99}$	0.01085505091630661248	−0.00058664697175542656
$r_{-0.997}$	−0.00332400849004544128	0.00018190473157963944
$r_{-0.999}$	0.00080162790195314304	−0.00004404312661758802
$r_{0.5}$	−0.59570526432450953216	0.02359068355075681792
$r_{0.8}$	6.317724918336749568	−0.11130490331799363584
$r_{0.9}$	−0.91064002464106315776	0.1851875415862827008
$r_{0.95}$	0.1246914619938220032	−0.31603770077908758528
$r_{0.98}$	−0.02733674892402686464	0.7781373339307139072
$r_{0.99}$	0.03200191266073124352	−1.86959847645702324224
$r_{0.993}$	−0.03240733988392482304	2.33024249274203045888
$r_{0.995}$	0.01317838979793997568	−1.08782871757080346624
$r_{0.998}$	−0.00147362112523202912	0.1500871954634144768
$r_{0.999}$	0.00050936617831053272	−0.05579478867087664128
$r_{0.9995}$	−0.0000965237331437649	0.01097937219909572864
a_1	−16.953724701048168775 68	−0.66422961761016676352
b_1	−5.78785780818708267008	0.616096068990383718 4
a_2	−0.36303650278996934656	−0.04795329185400025088
b_2	−0.30552601608781172736	0.03427398190138144256
a_3	0.03125802982031076864	0.00231314002746371744
b_3	−0.02225618146994111488	0.00225913379950247 52
a_4	−0.00456145412186675328	−0.00012050500396872546
b_4	−0.00463456822134962112	−0.00028595963389202888
a_5	0.00028030381010680324	0.00001591931476993 68
b_5	0.00083430654891606448	0.00004344320670827117
a_6	24.07539017048291016704	−2.402734876039174553 6
b_6	−25.17610778234234863616	−1.36886956837019631616
a_7	2.88263935918543437824	0.06824381050889828352
b_7	−0.23781397841299161088	−0.41023344111649021952
a_8	0.0151301186170249984	0.02973385974502390272
b_8	0.2155803876474701824	0.037696450928630451 2
a_9	−0.0141172945035984896	−0.00498885710405000768
b_9	−0.00984812725335878016	−0.00340748679269068 16
a_{10}	0.00199142789061523296	0.00073497323634501984
b_{10}	0.00070336128813542544	0.00041698896616485336
a_{11}	−0.082069076088726016	−0.00362543455216708608
b_{11}	−0.01734864938405026048	0.00597856049714365056
a_{12}	0.02151086124065270272	−0.00156085462037139808
b_{12}	0.00066041720651131496	−0.00205827276669462528
a_{13}	−0.0068158669788948096	0.00078386949448646256
b_{13}	−0.00037467928798129664	0.00052915082993023344
a_{14}	0.00067348282772420144	−0.00010203025958963884
b_{14}	0.00006539361844636485	−0.00004323615597633036

coefficient	p	v
a_{15}	20.0764581757059334144	5.1743060570980745216
b_{15}	89.1656915938120302592	−0.35398012652368232448
a_{16}	1.41684070718424629248	−0.5639169227195891712
b_{16}	−10.21895062224809754624	−0.55800346649178988544
a_{17}	−0.14255980830595647488	−0.07026661114511917056
b_{17}	0.82484463962066272256	−0.02947723163128034304
a_{18}	0.00612731805054674432	0.0004217314895111316
b_{18}	−0.07766393732289358848	−0.06904815394072582144
a_{19}	−0.00009338166677637412	−0.0025051638027132752
b_{19}	0.00961429993847092608	−0.02473498658638498816

Appendix C. Scripts and runs

All scripts are written in Python and take no arguments; their settings are constants at the top of each file. Except `measure_scout.py` and `inv_probe.py` (Section 5), which measure without pass criteria, each script states its pass criteria before running and ends by printing the number of failed criteria, and each of their runs listed here ended with zero failures. All runs were made by the author on his own computer on 7 and 8 October 2026.

script	content	log
`cat_cert4_new.py`	condition (C3): matrix B, $\zeta(2)$ contact, pivots modulo 131; positive control [1, Table 1] modulo 101	`_20261007_175948`
`cat_tail_edges.py`	trial coefficients of Proposition 15 (bases at the support edges; convex optimisation, rounding)	`_20261007_190940`
`cat_tail_opt.py`	coefficients of the 46-coefficient certificate (Remark 16)	`_20261007_172735`
`cat_cert7_new.py`	condition (C4): certificates for both coefficient files	`_20261007_191007`
`cat_newdesign_check.py`	structural conditions, $\zeta(2)$ contact, ranks modulo 131, valuations of Δ_1	`_20261007_165402`
`cat_design_search.py`	design search (Observation in Section 4.5)	`_20261007_162555`
`cat_equilibrium.py`	equilibrium problem (Section 6)	`_20261007_183129`
`sun_catalan_check.py`	Appendix D: [6, Proposition 3.1] at $B = 20$ and Observation 53	`_20261008_165354`
`measure_scout.py`	Observation 37, ratio $48 : 7 : 4 : 4 : 2$	`_20261008_183830`
`measure_scout.py`	Observation 37, ratio $96 : 15 : 6 : 8 : 5$, $n \leq 144$ and $n = 192$	`_20261008_192255`, `_220701`
`inv_probe.py`	Observation 40, $n = 96, 120, 144$	`_20261009_105215`

The scripts `cat_cert4.py`, `cat_cert7.py`, `cat_padic.py` and `cat_extend.py` of [3] check the preprint's own design. The remaining values of Observations 40 and 42 come from the scripts in `exponent/structure_scouts/`, run at sizes that finish in under a minute: `inv_probe.py` with $n = 24, 48, 72$ and, with the design as an argument, `sv_probe.py`, `gcol_probe.py` and `lam_probe.py` for the singular values, $u^{\mathsf{T}}Bv$ and the zeros at $n = 24, 48$, and `theta_probe2.py` and `alt_probe.py` for $-F'(1)$, r_{row} and r_{col}.

Appendix D. The preprint of Sun

This appendix gives the check of [6] mentioned in the introduction. The script and its run are listed in Appendix C.

The preprint [6] states that G is irrational. Wachs [7] shows that the estimate of [6, Theorem 9.1] does not follow from its derivation: with the worst-case denominator, the quantity controlled there lies between $+1.49$ and $+1.79$ per B^2 at eighteen indices $20 \le B \le 119$, and the prime 2 contributes $(2\log 2 + o(1))B^2$, which the derivation in [6, Secs. 6–9] does not carry. We recall the construction in the notation of [6] and give an independent exact computation of the same quantity, with the least common denominator of a polynomial in place of the worst-case denominator (Lemma 52).

Definition 51. For integers $B > S \ge 1$ put $N = 2B + S + 3$, $\Pi_i = \prod_{h=1}^{B} (2(h+i)+1)^2$ and $F_B = \prod_{r=0}^{2B-1} r!$. For an indeterminate X and $m \ge 0$ let

$$T_m(X) = (-1)^m \Big(X - \sum_{k<m} \frac{(-1)^k}{(2k+1)^2} \Big), \qquad u_m(X) = \frac{T_m(X)}{2m+1}.$$

For $0 \le a \le S+2$ and $1 \le j \le S$ let

$$R_{a,j}(X) = \sum_{i=0}^{a+2B} (-1)^i \binom{a+2B}{i} \Pi_i \, u_{i+j}(X),$$

and for $A \subset \{0, \dots, S+2\}$ with $|A| = S$ let

$$\hat{q}_{B,A}(X) = \frac{F_B \, \det\big(R_{a,j}(X)\big)_{a \in A,\, 1 \le j \le S}}{\prod_{i=0}^{N-1} \Pi_i} \in \mathbb{Q}[X],$$

a polynomial of degree at most S. Let $D_{B,A}$ be the least common denominator of its coefficients.

At $X = G$, $T_m(G) = (-1)^m \sum_{k \ge m} (-1)^k (2k+1)^{-2}$ is the tail of [6, (1.1)], and $\hat{q}_{B,A}(G)$ is the scalar $\hat{q}_B$ of [6, (3.5)]; we checked the identity [6, Proposition 3.1], which defines $\hat{q}_B$ through an $N \times N$ determinant, exactly at $B = 20$. Under the hypothesis $G = a/q$ with coprime integers a and $q \ge 1$, [6, Definition 3.1] lets $H_B^{\min}$ be the denominator of $q^S \hat{q}_B(a/q)$, and [6, Theorem 9.1] states that, for $S = \lfloor B/20 \rfloor$,

$$\log H_B^{\min} + \log |\hat{q}_B(G)| \le -\delta_0 B^2 + o(B^2), \qquad \delta_0 > 0.00966.$$

The valuation bounds in its proof [6, Lemma 5.3] use of a/q only that $qT_m(a/q)$ has the denominator of the partial sum; they hold for all integers a and $q \ge 1$. A bound for $H_B^{\min}$ that holds for all such a and q is at least $D_{B,A}/S!$, by the following lemma with $q = 1$.

Lemma 52. *Let $P \in \mathbb{Q}[X]$ be nonzero of degree at most S, and let D be the least common denominator of its coefficients. There is an integer a such that the denominator of $P(a)$ is at least $D/S!$.*

Proof. Write $DP = gP_0$ with g a positive integer and $P_0 \in \mathbb{Z}[X]$ primitive. Since D is the least common denominator, no prime divides both g and D. Write $P_0 = \sum_{k \le S} d_k \binom{X}{k}$; the $d_k = \Delta^k P_0(0)$ are integers, and the greatest common divisor d of the values $P_0(a)$, $a \in \mathbb{Z}$, equals that of the d_k. Since $S! \binom{X}{k} \in \mathbb{Z}[X]$ for $k \le S$, the coefficients of $S!\,P_0$ are integer combinations of the d_k, so d divides $S!$ times the content of P_0, which is $S!$. For each prime $p \mid D$ choose $a_p \in \mathbb{Z}$ with $v_p(P_0(a_p)) = v_p(d) \le v_p(S!)$; since $v_p(P_0(a))$ depends only on a modulo $p^{v_p(d)+1}$ when it is at most $v_p(d)$, the Chinese remainder theorem gives one integer a with $v_p(P_0(a)) \le v_p(S!)$ for every $p \mid D$. Then $v_p(P(a)) = v_p(P_0(a)) - v_p(D) \le v_p(S!) - v_p(D)$ for every $p \mid D$, and the denominator of $P(a)$ is at least $\prod_{p \mid D} p^{v_p(D) - v_p(S!)} \ge D/S!$. □

Since $\log S! = O(B \log B)$, the estimate of [6, Theorem 9.1], obtained from bounds of this kind, would give $\log D_{B,A} + \log|\hat{q}_{B,A}(G)| \le -\delta_0 B^2 + o(B^2)$ for the set A used there.

Observation 53 (Measured; exact arithmetic)**.** With $S = \lfloor B/20 \rfloor$, the minimum over the sets A with $\hat{q}_{B,A} \neq 0$ of $(\log D_{B,A} + \log|\hat{q}_{B,A}(G)|)/B^2$ is

B	20	40	60	80	100
minimum over A	$+1.612$	$+1.712$	$+1.752$	$+1.776$	$+1.789$

and the maximum over A exceeds the minimum by less than 0.03 in each case.

The measured values are positive and increase with B, while the estimate of [6, Theorem 9.1] requires them to tend to a limit at most -0.00966; they agree with the values of [7]. We do not rely on [6]. The acknowledgments of [6] state that the proof was developed in conversations with AI.